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# ATA-05

## TIME LIMITS/MAINTENANCE CHECKS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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## 05-00-00 Time Limits and Maintenance Checks

### General

The Maintenance Requirements Manual contains the scheduled maintenance tasks performed on the DHC-8-400 aircraft. Unscheduled maintenance tasks are included in the Aircraft Maintenance Manual Part 2, Chapter 05. Refer following sections for airworthiness limitations:

- For engine airworthiness limitation items refer to section 05-11-00.
- For Propeller airworthiness limitation items refer to section 05-13-00.
- For APU airworthiness limitation items refer to section 05-15-00.

### Engine Time Limits

The engine Time Limits, Operating Limits and Leading Particulars contain the limits approved by Transport Canada.

The operator must monitor the time limited items. The time limited items are identified in Pratt & Whitney Canada's Airworthiness Limitations Manual, Part No.: 3043520 for the PW150A (Build Spec. 885).

### NOTE

Hours/Engine Hours means Engine Flight Hours. Engine flight hour is defined as the engine operating time between aircraft takeoff and landing

### The use of restricted fuels

The restricted fuels, as listed in the Airplane Flight Manual (PSM 1-84-1A), are considered by P&WC to be satisfactory for occasional use only provided they are not used in an engine for more than a cumulative total of 1000 hours.

The 1000 hours can be accumulated intermittently or continuously. For intermittent use of restricted fuels, start accumulating time as soon as the aircraft is refuelled with restricted fuel, regardless of the quantity added. Stop accumulating time as soon as the aircraft is refuelled with a quantity of approved fuel that is at least 70% of the fuel tank capacity.

When a total of 1000 hours of engine running has been accumulated with restricted fuels, a fuel nozzle inspection is recommended. P&WC recommends sending the fuel nozzles to an approved repair facility to be tested as received.

## Example of Engine Time Limits:

### Over Torque

AREA A: 1. Allowable time-limited torque.  
2. No unscheduled maintenance action required.

AREA B: 1. Investigate and correct cause of overtorque.  
2. Carry out maintenance actions specified in Task 05-53-00-210-805

AREA C: 1. Investigate and correct cause of overtorque.  
2. Carry out maintenance actions specified in Task 05-53-00-210-805

AREA D: 1. Investigate and correct cause of overtorque.  
2. Remove reduction gearbox (RGB) module and return to an overhaul facility for inspection.

NOTE: Magnitude and duration of torque in areas B,C & D must be recorded in turbomachinery/reduction gear log book.

NOTE: 100% torque = 26113 lb.ft

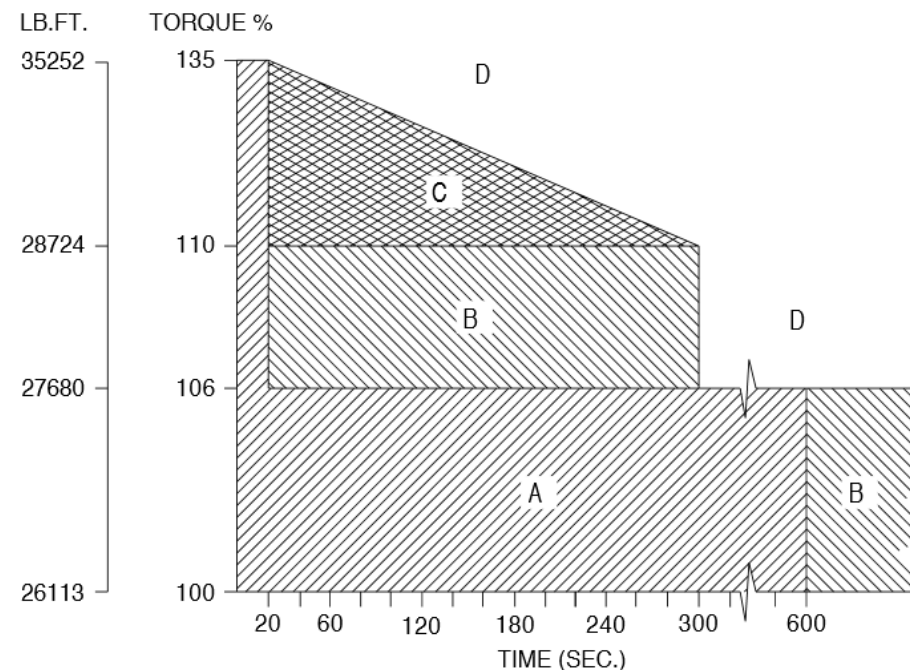


Figure 05-01 Engine- Over torque Limits

## Over Temperature

**AREA A:** NO ACTION REQUIRED IF CONDITION OCCURS DURING START (SEE NOTE 1). OTHERWISE DETERMINE AND CORRECT CAUSE OF OVERTEMPERATURE. RECORD IN ENGINE LOG BOOK.

(1) NO ACTION REQUIRED IF CONDITION OCCURS DURING TRANSIENT (SEE NOTE 2).

(2) IF CONDITION OCCURS DURING STEADY STATE, DETERMINE AND CORRECT CAUSE OF OVERTEMPERATURE. RECORD IN ENGINE LOG BOOK.

**AREA B:** (1) INVESTIGATE AND CORRECT CAUSE OF OVERTEMPERATURE.  
(2) REMOVE TURBOMACHINERY MODULE AND RETURN TO AN OVERHAUL FACILITY FOR AN INSPECTION AS IT IS WRITTEN IN THE PRATT & WHITNEY CANADA PW150A ENGINE MANUAL.

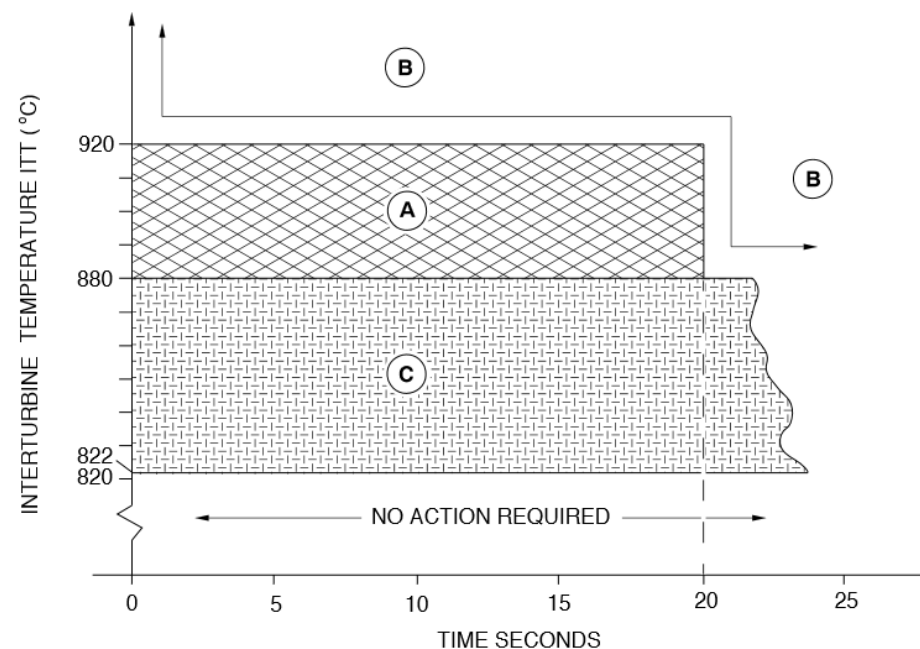
**AREA C:** INVESTIGATE AND CORRECT CAUSE OF OVERTEMPERATURE (SEE NOTES 3 AND 4).

**NOTE 1:** TROUBLE SHOOTING IS RECOMMENDED (REF. ENGINE FAULT ISOLATION) WHEN TEMPERATURE IS CONSISTENTLY IN AREA "A".

**NOTE 2:** TRANSIENT IS DEFINED AS CHANGE IN TEMPERATURE CAUSED BY MOVE-MENT OF THE POWER OR CONDITION LEVER OR AN ALTITUDE CHANGE.

**NOTE 3:** AREA "C" REPRESENTS THE NORMAL TAKE OFF (NTO) LIMIT, ADJUSTED FOR AMBIENT CONDITIONS. AN EXCEEDANCE OCCURRED ONLY IF THE TEMPERATURE INDICATE "RED" ON THE COCKPIT GAUGE.

**NOTE 4:** NO ACTION REQUIRED IF CONDITION OCCURS DURING START OR TRANSIENT.



**Figure 05-02 Engine– Over Temperature Limits**

## 05-50-00 Unscheduled Maintenance Checks

ATA chapter 05 of the AMM contains unscheduled maintenance tasks that must be completed after non-normal operations have occurred before the aircraft may be returned to service.

### 05-50 Unscheduled Maintenance Checks

- Hard Landing
- Tail Strike
- Dangerous Turbulence or Buffeting
- Bird Strike
- Wheel Bearing Failure
- Lightning Strike
- Operation at more than the Maneuvering Load Factors
- Operation at more than the Maximum Operating Speed
- Operation at more than the Maximum Gear Operation Speed
- Operation at more than the Wing Flaps Down Speed
- Aircraft Park in Strong Wind or Wind Gust
- Application of Too Much Ground Deicing Fluid
- APU After Hard Metal Ingestion
- Rough Towing
- Application of Maximum Braking
- Tire Explosion
- Volcanic Ash Condition
- Water Accumulation

#### NOTE

Other task may exist; check the AMM to ensure no tasks are missed.

## 05-53 Engine Unscheduled Maintenance Checks

- Engine inspection for Fuel in the Oil System
- Engine inspection After an Overspeed that is more than Transient
- Engine inspection After Overtemperature
- Engine inspection After Overtorque
- Engine inspection After Immersion in Water
- Engine inspection After Hard Metal Ingestion
- Engine inspection After Chip Detector Circuit Completed
- Engine inspection After Propeller Sudden Stoppage
- Engine inspection After Propeller Strike Causing Minor Blade Damage
- Engine inspection After Low Oil Pressure or Loss of Oil Pressure
- Engine inspection After Oil Fire in exhaust Duct
- Engine inspection After Engine Exposed to Fire Extinguishing Agent
- Engine inspection After AC Generator Failure
- Engine inspection After an Overspeed Governor Shaft Fracture
- Propeller inspection After Bird Strike
- Propeller inspection After Lightning Strike
- Propeller inspection After an Overspeed Condition
- Propeller inspection After an Overtorque
- Engine inspection After Storage Without Preservation
- Engine inspection After Propeller Lightning Strike
- Engine inspection After Hydraulic Fluid in the Fuel System
- Fuel Nozzle Inspection After Use of Restricted Fuel

#### NOTE

Other task may exist; check the AMM to ensure no tasks are missed.

**05-54 Hydraulic System Unscheduled Maintenance Checks**

- Inspection of the Hydraulic Engine Driven Pump EDP after Operation without Sufficient Hydraulic Fluid
- Inspection of the Power Transfer Unit (PTU) Due to Shaft Seal Leakage

**NOTE**

Other task may exist; check the AMM to ensure no tasks are missed.

**05-61 Propeller System Unscheduled Maintenance Checks**

- Propeller inspection After an Overspeed Condition
- Propeller inspection After an Overtorque
- Propeller Inspection for Excessive Grease Leakage and/ or Signs of Loose Blade
- Propeller Inspection After a Bird Strike
- Propeller Inspection After a Lightning Strike
- Propeller Inspection After an Engine Fire
- Propeller Inspection for Signs of Crosshead Shaft Seal Failure/Leakage
- Propeller Inspection After Operation in High Crosswind

**NOTE**

Other task may exist; check the AMM to ensure no tasks are missed.





# ATA 06

## DIMENSIONS AND AREA

## 06-10-00 Dimensions

### General

Aircraft Dimensions are:

### Overall Dimensions

Span 93 ft 3 in (28.42m)

Length 107 ft 9 in (32.83 m)

Height 27 ft 4 in (8.34 m)

Propeller Ground Clearance 38.67 in (98.22)

### Wings

Root chord 121.26 in (3.08 m)

Tip chord 46.59 in (1.18 m)

Aspect ratio 12.8

### Horizontal Stabilizers and Elevators

Span 30 ft 5 in (28.42 m)

Root chord 6 ft 8 in (2.03)

Tip chord 4 ft 11 in (1.50 m)

Aspect ratio 2.27

Incidence 1 degree

Dihedral 0 degrees

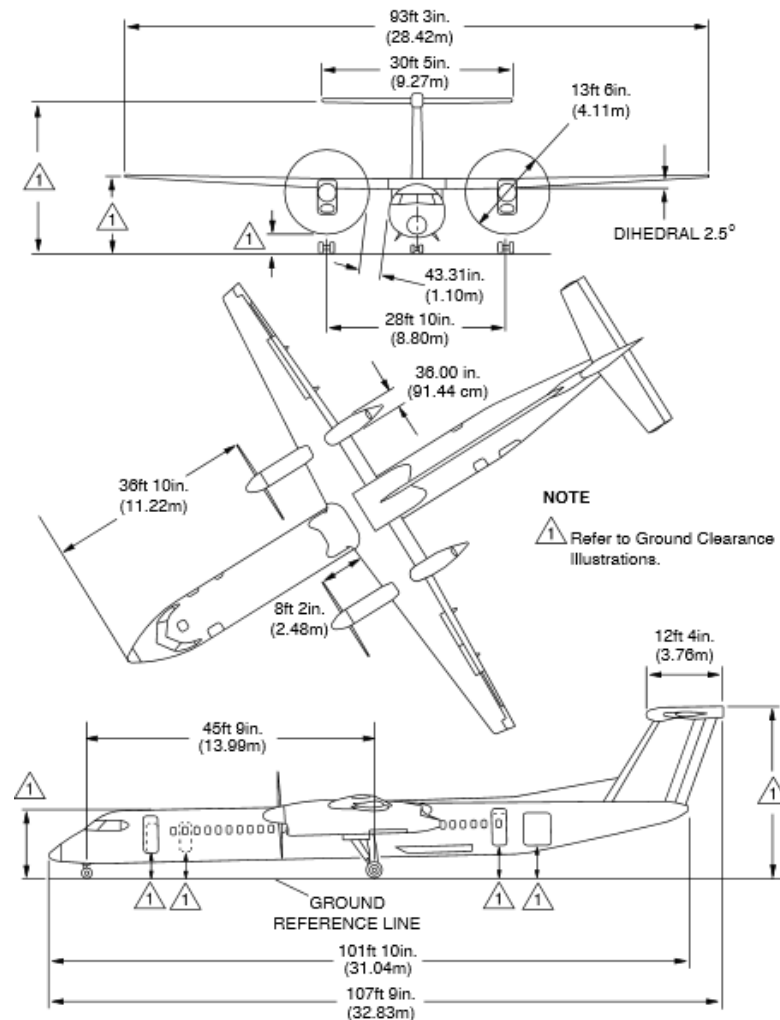


Figure 06-01 Aircraft External Dimensions

## 06-20-00 Aircraft Stations

### General

#### Aircraft Axis

The three primary axis are given X, Y and Z and the ordinates are identified along these axis from a point of origin. The longitude axis is X, the lateral axis is Y and the vertical axis is Z. The point of origin for the three axes is in front of the forward passenger door, centre line and below the ground reference line of the aircraft. Identification of the ordinates is by the dimension in inches from the point of origin. The X, Y, Z axis system has more points of origin that are set in the aircraft. These datum points show the location of primary assemblies.

The wings, horizontal stabilizer, nacelles and vertical stabilizer are primary assemblies.

These ordinates are identified by:

- $X_W$ ,  $Y_W$  and  $Z_W$  (Wings)
- $X_H$ ,  $Y_H$ , and  $Z_H$  (Horizontal stabilizer)
- $X_V$ ,  $Y_V$ , and  $Z_V$  (Vertical stabilizer)
- $X_N$ ,  $Y_N$ , and  $Z_N$  (Nacelles)
- $X_P$ ,  $Y_P$  and  $Z_P$  (Bullet fairing).

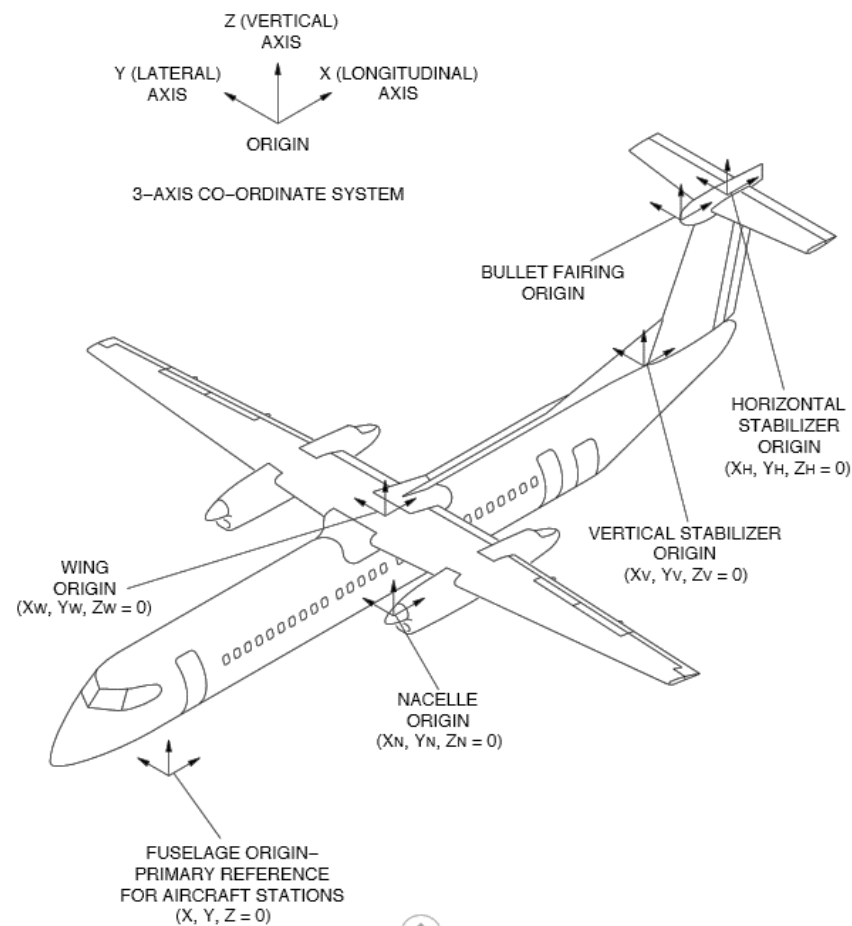


Figure 06-02 Axis of Aircraft

## 06-30-00 Aircraft Zones

### Refer Figure 06-3: Major Zones

The aircraft is divided into eight major zones and these are:

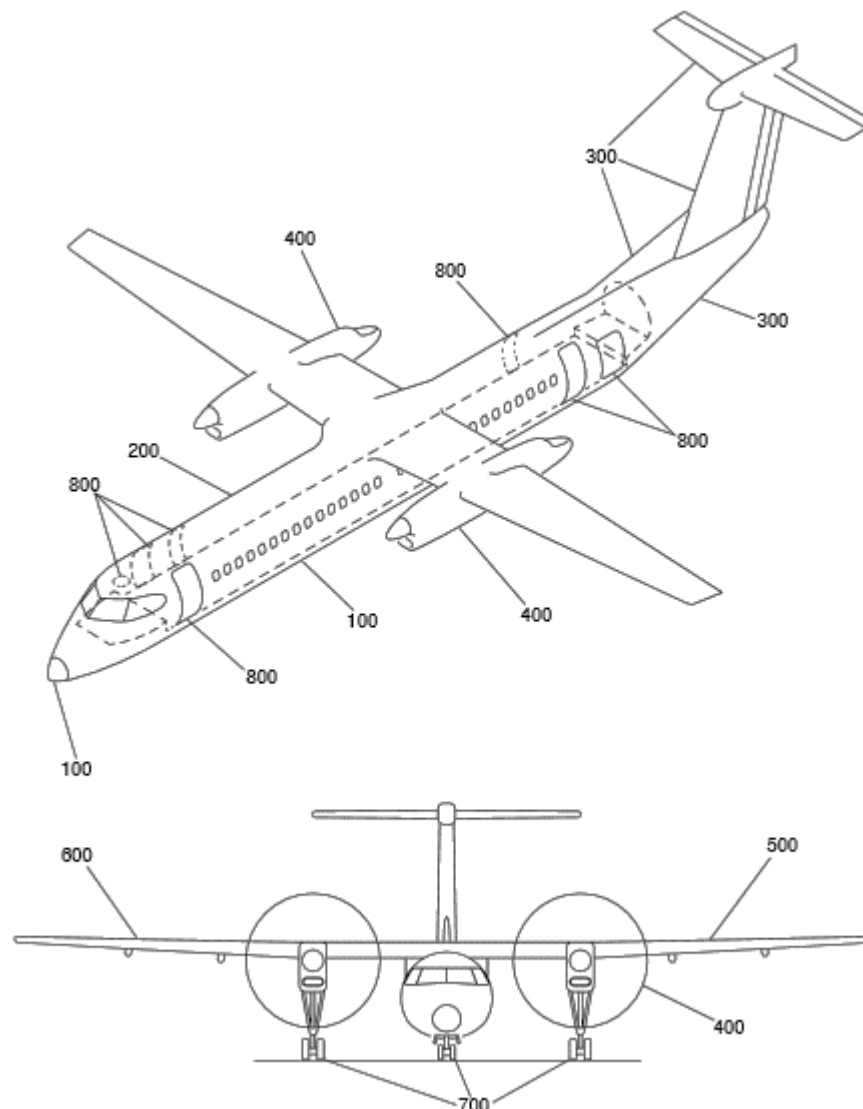
- Zone 100 Lower fuselage
- Zone 200 Upper fuselage
- Zone 300 Empennage
- Zone 400 Powerplants and nacelles
- Zone 500 Left wing
- Zone 600 Right wing
- Zone 700 Landing gear and landing gear doors
- Zone 800 Doors

Each major zone is further broken down into subzones for ease of location components and panels.

Some examples of these are:

Subzone 120 Flight compartment under floor area

Subzone 130 Fuselage under floor area



**Figure 06-03 Zones of Aircraft**

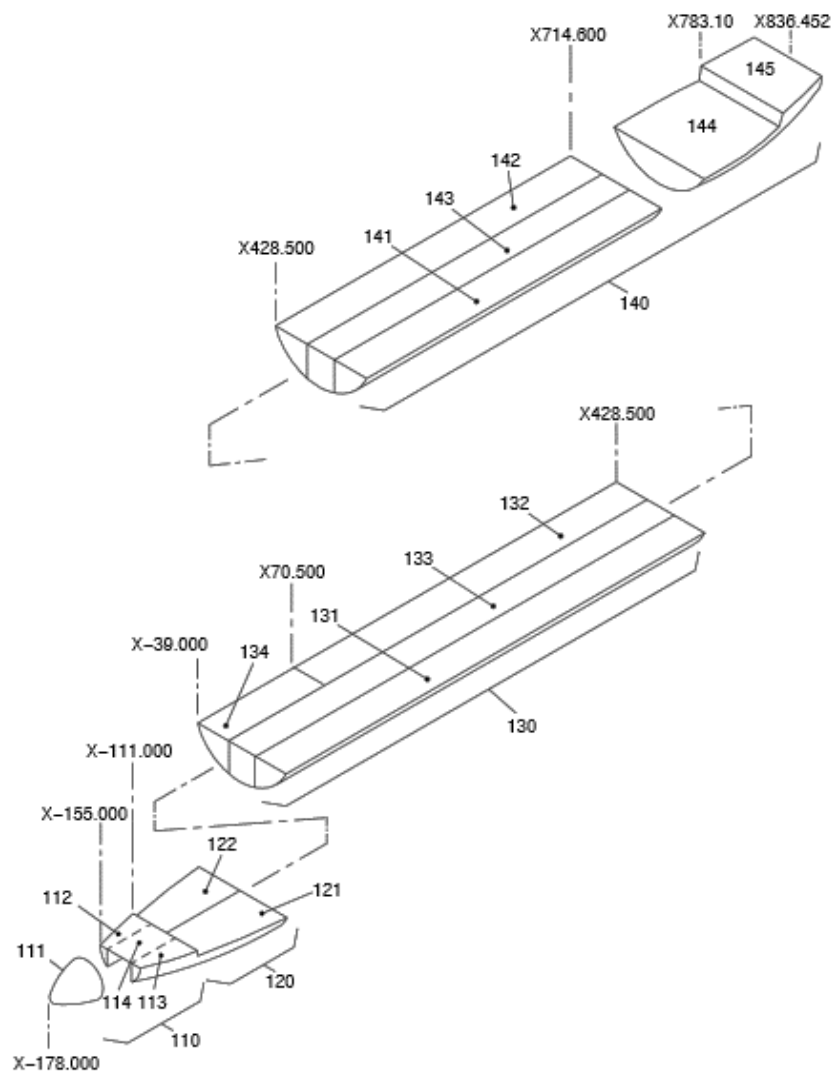


Figure 06-04 Sub Zones-Lower Fuselage

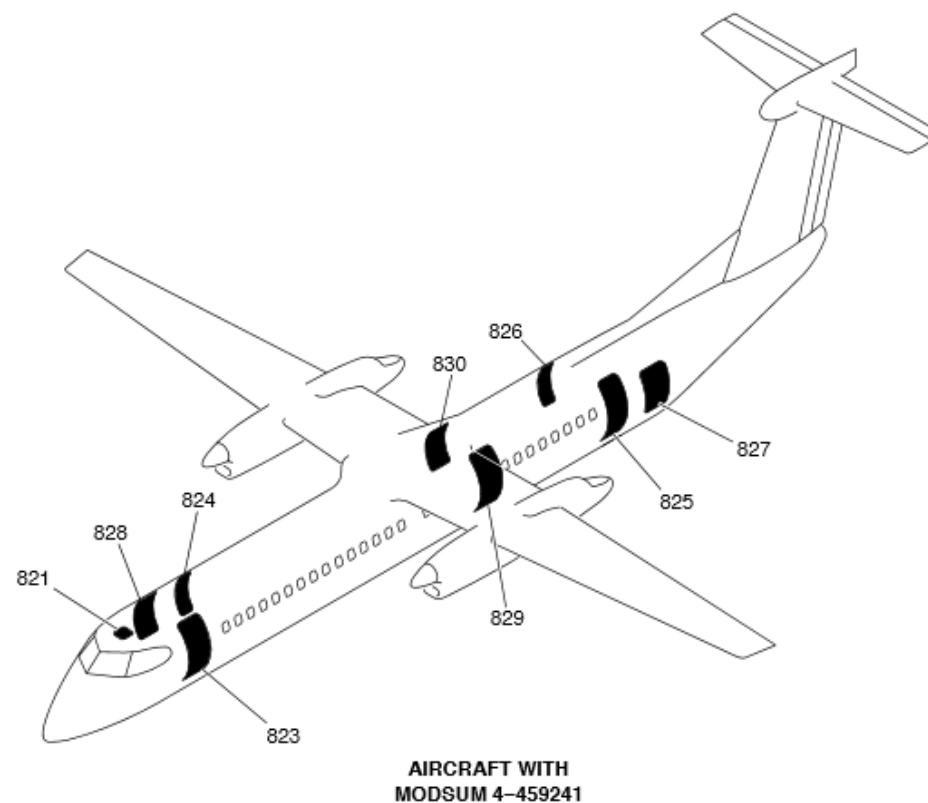


Figure 06-04 Sub Zones – Fuselage Doors

## 06-41-00 External Access Panels and Doors

### Refer Figure 06-6:

Access Panels and Doors The access panels and doors are identified by an alphanumeric code. The identification code makes it easier to find the different access panels, doors and fairings. Each panel, door and fairing has a two-group code to indicate its position on the aircraft. A three digit numeric group indicates the Major zone and sub zone where the panel, door or fairing is located. A two digit letter group gives the position of the panel, door or fairing in relation to the aircraft as follows:

- The first letter identifies the general position. This starts with the letter A and goes from the inboard to outboard and from the front to the rear.
- The second letter identifies the location of the panel, door or fairing in relation to the aircraft as follows:
  - T (Top)
  - B (Bottom)
  - L (Left side)
  - R (Right side)
  - Z (Internal)

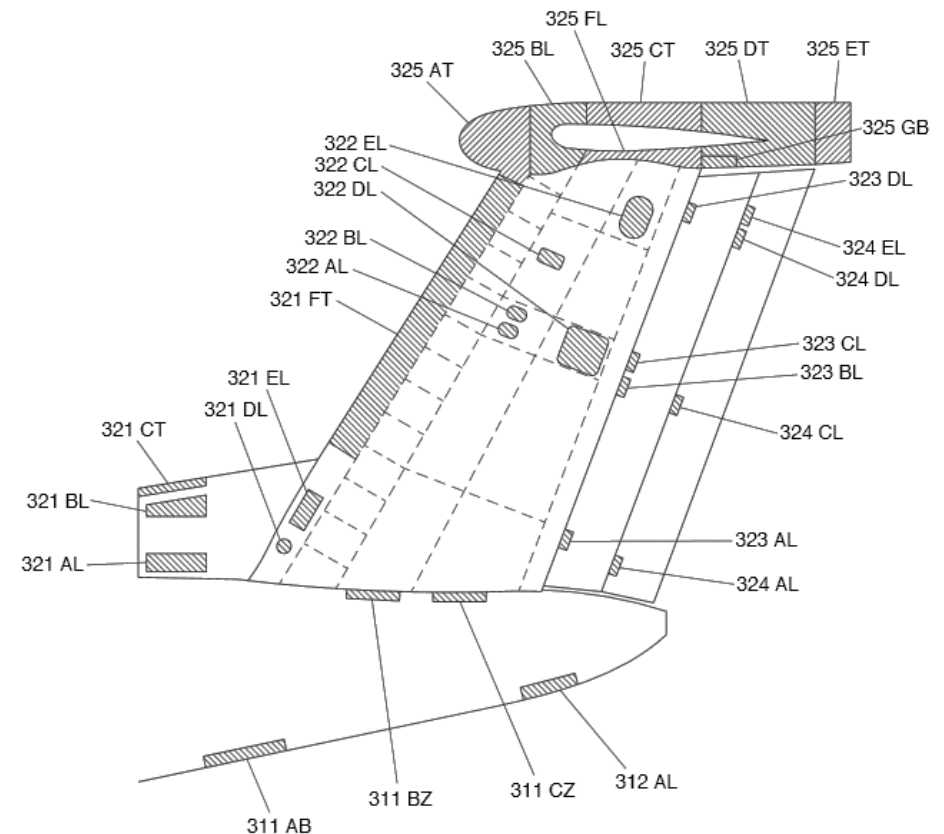


Figure 06-05 Sub Zones-Lower Fuselage



# ATA-07

## LIFTING AND SHORING



## 07-00-00 Lifting and Shoring

### Introduction

There are three jacking points on the aircraft structure for jacking of the aircraft. There are two jacking points on the rear spar on the inboard area of the left and the right wings. There is one jacking point on the forward fuselage, aft of the nose landing gear (NLG).

There is a jacking point for jacking of the aircraft at the nose landing gear axle. There is a jacking point for jacking of the aircraft at each of the two the main landing gear axles.

### General

Refer to Figure 07-1: Jacking of the Aircraft

If necessary, the complete aircraft can be lifted without payload (engines and propellers or equivalent weight installed). If necessary, the engines can be removed when the aircraft is on the jacks. Ballast is distributed equally on the floor of the cockpit / cabin area forward of the lavatory as follows:

- 500 lbs (227.3 kgs) of ballast if only one engine is to be removed
- 1000 lbs (454.6 kgs) of ballast if both engines are to be removed

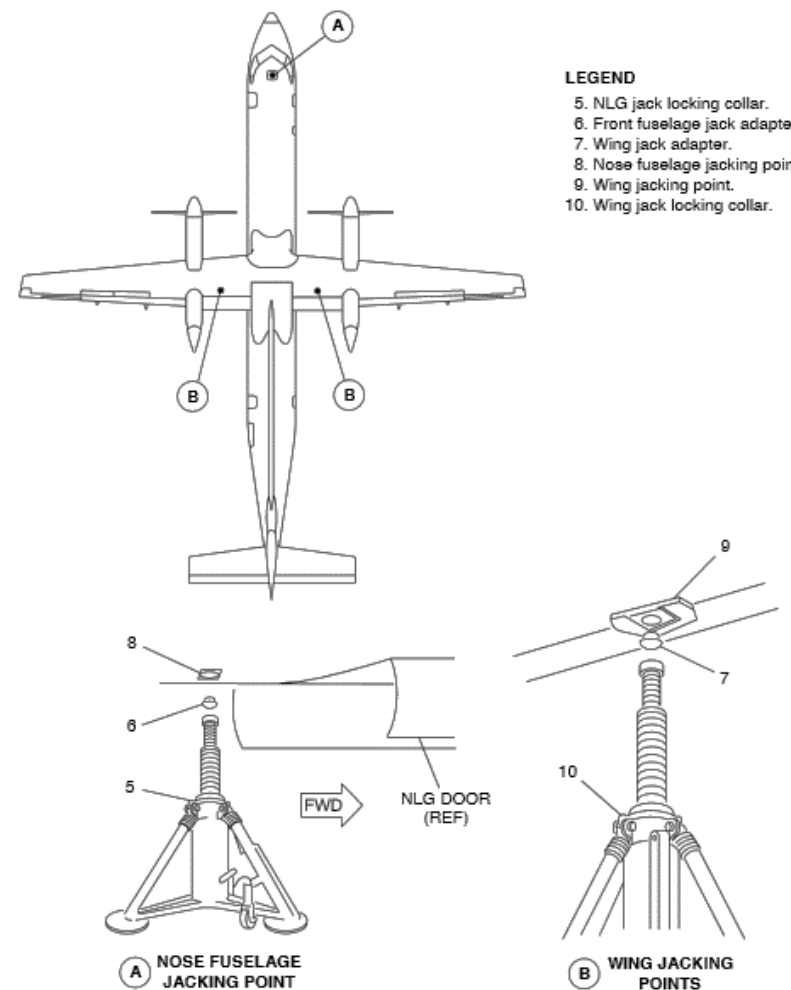


Figure 07-01 Aircraft Jacking Points



If necessary, the aircraft nose landing gear (NLG) can be lifted only using the structural jacking point aft of the NLG. The two main landing gears (MLG) must stay on the ground. The aircraft weight must be less than 48,792 lb (22,132 kg) and the areas above and below the aircraft are clear. The aircraft can be lifted with axle jacks for the nose gear or for the two main gears. The aircraft can be lifted at the nose fuselage and the main landing gear Axles for weighing the aircraft.

**CAUTION:** DO NOT LIFT THE AIRCRAFT IN AN OPEN AREA WIND SPEED IS MORE THAN 20 MPH (32 KM/H).

**WARNING:** DO NOT LIFT THE AIRCRAFT ON JACKS IF YOU THINK THE LOAD AT THE NOSE-FUSELAGE JACKING POINT WILL BE 4700 LB (2132 KG) OR MORE.

**WARNING:** DO NOT LIFT THE AIRCRAFT AT THE JACKING POINTS OF THE WING AND THE NOSE FUSELAGE IF THE WEIGHT IS MORE THAN 48,792 LB (22,132 KG).

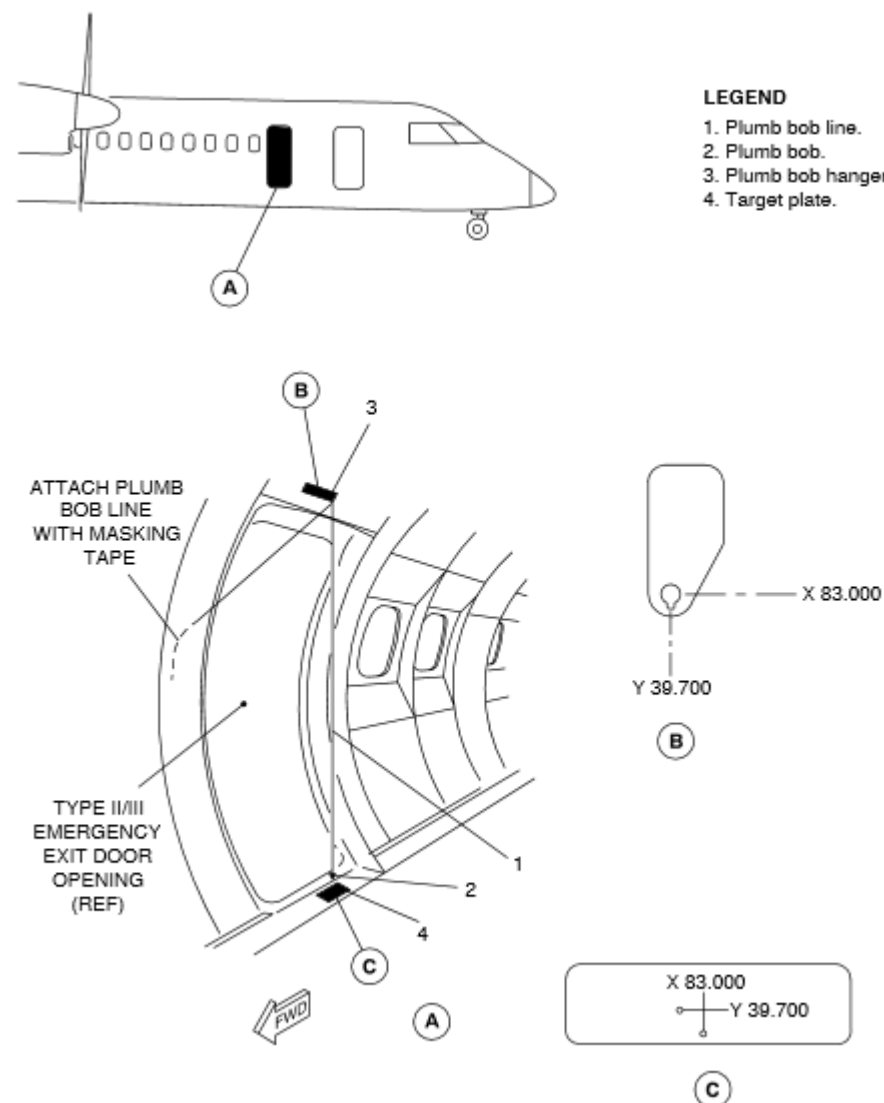
**WARNING:** BEFORE YOU LIFT THE AIRCRAFT ON JACKS, MAKE SURE THAT:

- NO PERSON IS IN/ON THE AIRCRAFT
- YOU RELEASE THE PARKING BRAKE
- YOU REMOVE THE CHOCKS FROM THE WHEELS
- THERE IS SUFFICIENT CLEARANCE AROUND/ABOVE THE AIRCRAFT.

IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT.

Before positioning the wing jacks, the aircraft shall be levelled by lifting the Nose Body jack.

Refer Figure 07-2 for levelling of 78 seater aircraft and figure 07-3 for 90 seater aircraft.



**Figure 07-02 Aircraft Levelling Point (78 seater)**

### Refer Figure 07-03: Levelling Extra Capacity Aircraft

In the Extra Capacity (90 seater aircraft) the plumb bob hanger is on the top surface of the Type I emergency – exit – door (written as “emergency exit door” in this procedure) on the right side of the fuselage. The target plate is on the bottom surface of the forward Type 1 emergency exit door. The center mark of the target plate is at Sta. X19.5 and Y41.5.

**WARNING:** MAKE SURE YOU DO NOT TURN THE DITCHING DAM HANDLE WHEN THE FORWARD EMERGENCY EXIT TYPE I DOOR OR TYPE II/III EMERGENCY EXIT IS OPEN. IF YOU DO THIS, THE DITCHING DAM IS RELEASED AND WILL FALL. THIS CAN CAUSE DAMAGE TO EQUIPMENT AND/OR INJURIES TO MAINTENANCE PERSONNEL

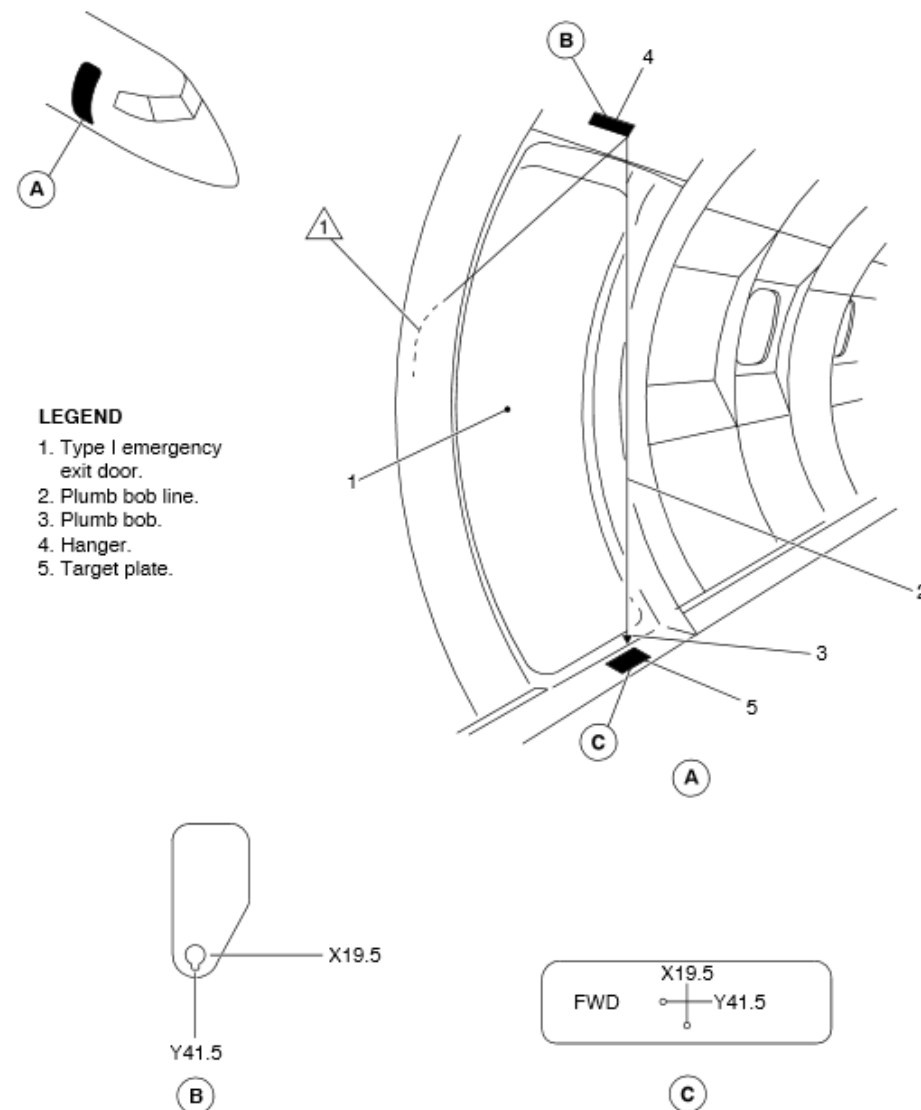


Figure 07-03 Aircraft Levelling Point (90 seater)

### Refer to Figure 07-04: Jacking of the Aircraft at the Main Landing Gear Axle

A jack at the MLG axle is used to lift the aircraft as follows:

- The MLG axle jack is put at the MLG axle jacking point
- The jack height is Increase until the jack correctly engages the jacking point
- The axle jack is used to lift the aircraft until the related wheels are appx 2.0 inches above the ground.

The aircraft is lowered as follows:

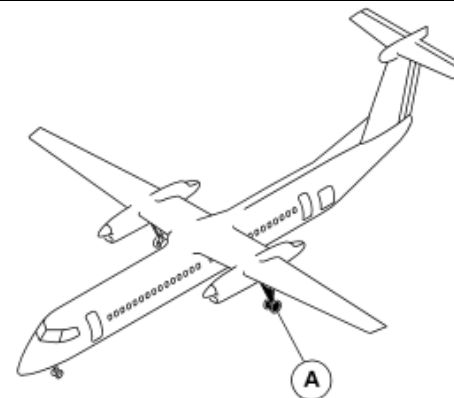
- Lower the aircraft until the MLG has weight on wheels
- Move the jack clear of the aircraft.

**WARNING:** BEFORE YOU LOWER THE AIRCRAFT ON JACKS, MAKE SURE THAT:

- NO PERSON IS IN/ON THE AIRCRAFT
- YOU RELEASE THE PARKING BRAKE
- YOU ENGAGE THE DOWN LOCK OF THE NOSE LANDING GEAR
- YOU INSTALL THE GROUND LOCKPINS OF THE MAIN LANDING GEAR
- THERE IS SUFFICIENT CLEARANCE AROUND/ABOVE THE AIRCRAFT

**WARNING:** MAKE SURE THAT YOU INSTALL THE MLG DOOR GROUND LOCK PINS. THE MLG DOORS WILL CLOSE WHEN YOU PRESSURIZE THE NO.2 MAIN HYDRAULIC SYSTEM IF THE MLG DOOR GROUND LOCK – PINS ARE NOT INSTALLED. IF THIS OCCURS, IT CAN CAUSE DAMAGE TO EQUIPMENT OR INJURIES TO PERSONNEL

**WARNING:** INSTALL LANDING GEAR GROUND LOCK PINS TO THE MAIN LANDING GEAR. ENGAGE THE NOSE LANDING GEAR GROUND LOCK



#### LEGEND

1. MLG axle jack.
2. MLG axle jacking point.

#### NOTE

Left MLG is shown, right MLG is similar.

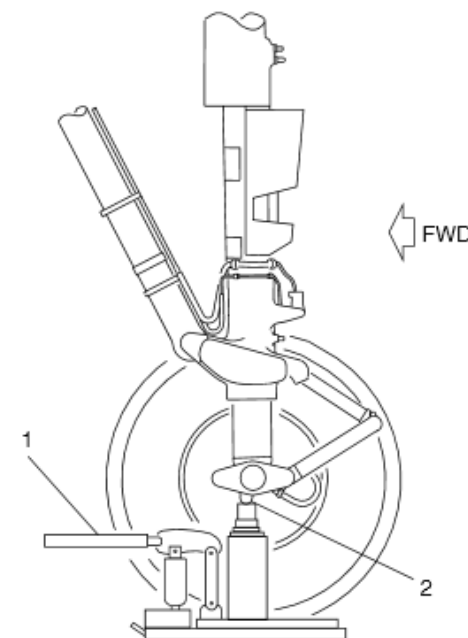


Figure 07-04 Axle Jacking Point (MLG)

### Refer to Figure 07-5: Jacking of the Aircraft at the Nose Landing Gear Axle

Lift the aircraft using a jack at the NLG axle as follows:

- (1) Center the nose wheel.
- (2) Make sure that the nose wheel STEERING is OFF.
- (3) Install the NLG castor lock pin in its hole.
- (4) Put the NLG axle jack (1) at the NLG jacking point (2).

NOTE: You can move a chock away from the wheel of the NLG to get access for the jack.

- (5) Increase the jack height until the jack correctly engages the jacking point
- (6) Lift the aircraft using the axle jack until the related wheel is approximately 2 in. (5.1 cm) clear of the ground. Make sure that you continuously move the lock collar down while you lift the aircraft.

Lower the aircraft using the jack (1) at the NLG axle as follows:

**WARNING:** BEFORE YOU LOWER THE AIRCRAFT THAT IS ON AN AXLE JACK, MAKE SURE THAT:

- YOU ENGAGE THE DOWN LOCK OF THE NOSE LANDING GEAR
- YOU INSTALL THE GROUND LOCKPINS OF THE MAIN LANDING GEAR
- THERE IS SUFFICIENT CLEARANCE ABOVE/BELOW AND AROUND THE AIRCRAFT.

IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND/OR DAMAGE TO THE AIRCRAFT.

- (1) Lower the aircraft using the jack until the NLG has weight on wheels.
- (2) Move the jack clear of the aircraft.

NOTE: If you moved a chock away from a wheel to get access then you must put the chock back.

- (3) Make sure that the nosewheel is in the center and STEERING s/w is OFF.

- (4) Remove the NLG castor lock pin from its hole

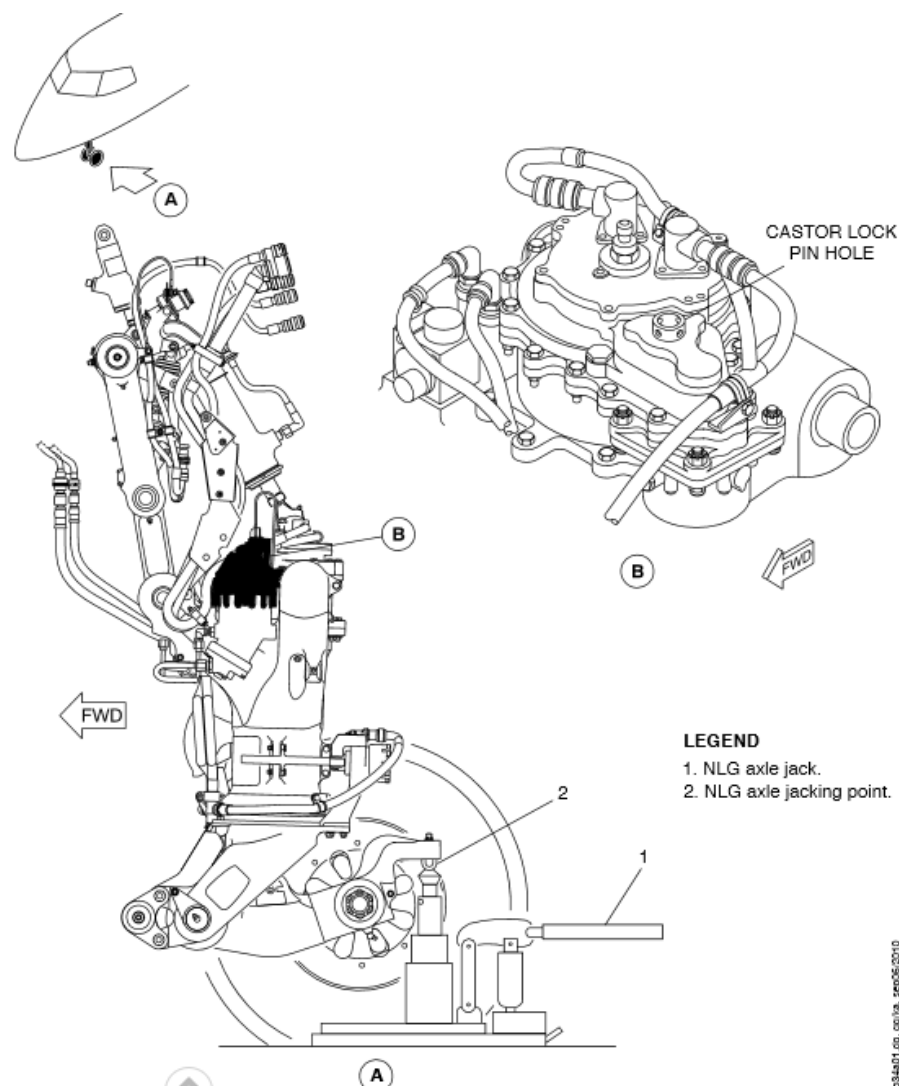


Figure 07-05 Axle Jacking Point (NLG)



# ATA-08

## LEVELLING & WEIGHING

## 08-00-00 Levelling & Weighing

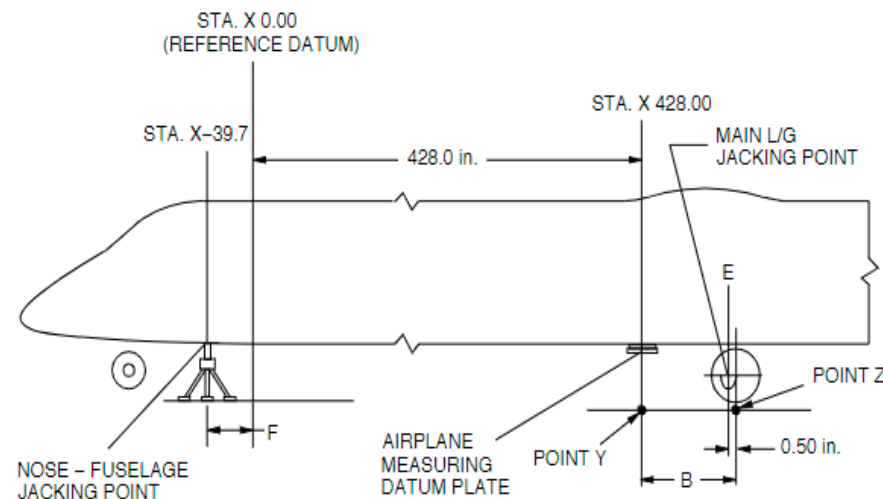
Electronic load sensing devices must be used to weigh the aircraft at the nose fuselage and the axles of the main landing gear. The weights are recorded to calculate the total weight and the center of gravity of the aircraft.

The basic weight of the aircraft is that weight that includes all fixed operating equipment (both standard and optional), trapped and unusable fuel, and full engine oil. You must put the aircraft in the basic weight configuration that follows before you weigh it:

- (1) Make sure the landing gear is extended. Make sure the MLG lockpins are installed and the nose-landing-gear ground lock is engaged.
- (2) Make sure the flaps are fully retracted.
- (3) Make sure the flight controls are in the neutral position with the gust lock applied.
- (4) Make sure all doors and panels are installed and closed (the forward right emergency exit door not included)
- (5) The aircraft must be in a level position.
- (6) Remove the food supplies from the galley. Do not remove emergency equipment kept in the galley.
- (7) Drain the galley liquid containers.
- (8) Make sure all emergency equipment is stowed in its correct location.
- (9) Drain the fuel left in the tanks and surge bays through their related drain valves on the bottom of the wings. Use the fuel sampler unit to do this. The remaining fuel in the tanks is trapped fuel and is part of the Basic Weight of the aircraft.
- (10) Make sure that the engine oil and all the hydraulic reservoirs are full.
- (11) Do the servicing of the lavatory. Full toilet fluid is part of the basic weight.
- (12) Drain the wash water.

(13) Remove all tools and equipment not included in the Basic Weight configuration (Refer to PSM 1-84-8, Weight and Balance Manual, Chapter 2)

(14) Park the aircraft in a closed hangar with the fans off. Make sure that the temperature in the hangar is stable.



### NOTES

"F" (Horizontal Nose-Fuselage Reaction) = -39.7 inches.

Distance from Reference Datum to "E" (Main Reaction) =  $428 + B - 0.50$  ("B" is a measured value).

**Figure 08-01 Weight Reaction – Point Measurements**



**WARNING:** DO THE LEVELING AND WEIGHING PROCEDURE IN A CLOSED HANGAR. IF YOU DO NOT DO THIS, WIND CAN CAUSE INCORRECT RESULTS. THE AIRCRAFT MAY NOT BE IN A SAFE BALANCED CONDITION FOR TAKE OFF, FLIGHT, AND LANDING.

Weigh the aircraft on jacks with the electronic load sensing kit as per PSM 1-84-8, Weight and Balance Manual, Chapter 2.

**WARNING:** DO NOT USE THE WING JACKING POINTS TO LIFT THE AIRCRAFT WHEN YOU DO THE LEVELING AND WEIGHING PROCEDURE. IF YOU DO, YOU WILL CAUSE SIDE LOADS TO OCCUR. THIS WILL GIVE YOU INCORRECT INDICATIONS FROM THE LOAD-CELLS. THE AIRCRAFT MAY NOT BE IN A SAFE BALANCED CONDITION FOR TAKE OFF, FLIGHT, AND LANDING.

You must do the procedures that follow as you weigh the aircraft:

- Jacking of the Aircraft
- Leveling of the Aircraft

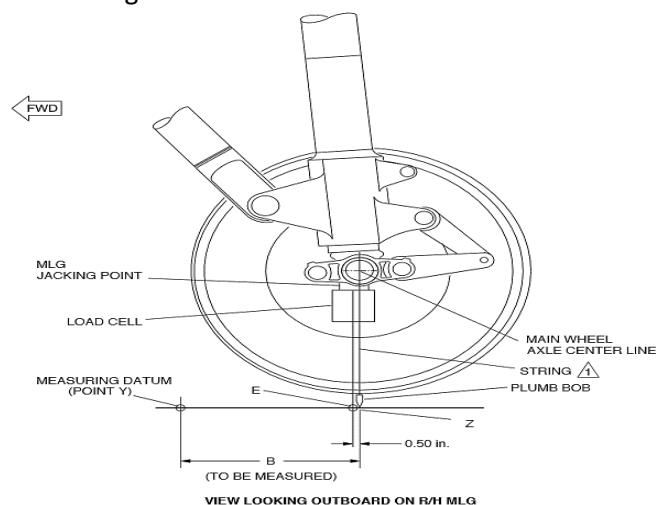
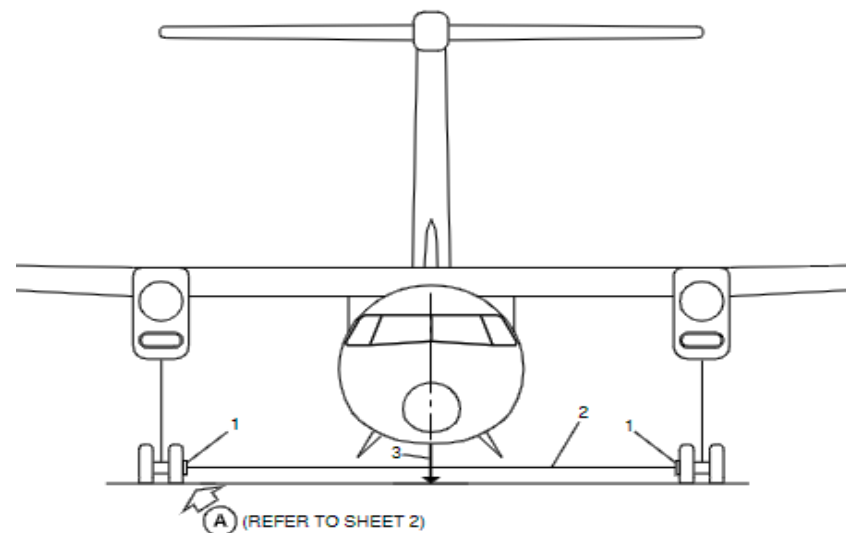


Figure 08-02 Measuring Data – Weighing on MLG Jack Points

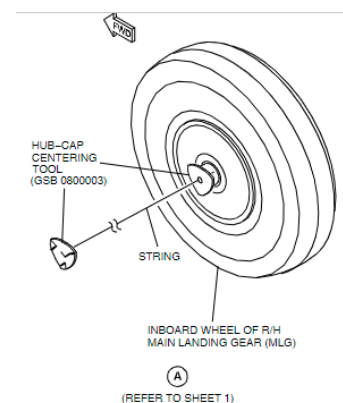


**LEGEND**

1. Hub-cap centering tools. (GSB 0800003)
2. String.
3. Plumb bob and string (ref.).

**NOTE**

Nose Landing Gear not shown for clarity.



How to use Hub-Cap Centering Tool to Determine the Center Line of Main Wheels:

1. Use masking tape to attach the centering tools to the hub caps of the inboard wheels.
2. Pull the string tight and secure it.
3. Drop the plumb bob from the center line of the fuselage bottom surface.
4. Mark the center line of the main.

Figure 08-03 Method to Determine the Center Line of Main Wheels



# ATA-09

## TOWING AND TAXIING



## 09-00-00 Towing and Taxiing

### 09-10-00 Towing

The aircraft is towed or pushed from the nose landing gear with a tow vehicle and a tow bar. A minimum of two crew members is necessary to tow the aircraft. One person operates the tow vehicle. The other person must be in the flight compartment to operate the parking brake in an emergency. The crew member in the flight compartment will also apply the parking brake when you park the aircraft. Make sure a crew member stays at each wing tip and at the tail when you push or tow the aircraft in an area with minimum space.

Tow the aircraft as follows:

- (1) Engage the ground lock of the nose landing gear and Install the lock pins in the main landing gear.
- (2) Do a pressure check of the parking brake system at the accumulator gauge in the right wing root. Make sure that the pressure is a minimum of 1000 psi.
- (3) If the pressure of the parking brake system is below 1000 psi, use the hand pump to pressurize the system.
- (4) If towing is to be done with one or both engines removed, make sure that ballast weights are loaded and uniformly distributed on the floor of the cockpit and forward cabin area forward of the lavatory as follows:
  - Load and distribute a minimum of 500 lbs (227 Kg) of the ballast weights if towing is done with one engine removed.
  - Load and distribute a minimum of 1000 lbs (454 Kg) of ballast weights if towing is done with both engines removed.

- (5) Disconnect the grounding wire from the aircraft.
- (6) Make sure that external servicing connections are removed from aircraft.
- (7) Do a general visual inspection of the landing gear tires to make sure that the tires are serviceable.
- (8) Close all external doors of the aircraft.

**CAUTION:** OBEY THE PRECAUTIONS THAT FOLLOW WHEN YOU TOW THE AIRCRAFT:

- DO NOT TOW THE AIRCRAFT AT MORE THAN 5 MPH (8 KPH) AROUND CORNERS
- DO NOT TOW THE AIRCRAFT AT MORE THAN 15 MPH (24 KPH) IN A STRAIGHT LINE
- HAVE A PERSON IN THE FLIGHT COMPARTMENT TO APPLY BRAKES IN AN EMERGENCY
- DO NOT USE THE EMERGENCY/PARKING BRAKES TO STOP THE AIRCRAFT UNLESS THERE IS AN EMERGENCY
- IF THERE IS AN EMERGENCY STOP, AN INSPECTION IS NECESSARY – INCREASE OR DECREASE SPEED SMOOTHLY AND GRADUALLY
- MAKE TURNS LARGE WHEN POSSIBLE
- MAKE ALL CHANGES TO SPEEDS OR DIRECTIONS SLOWLY
- WHEN YOU STOP THE TOW VEHICLE, DO NOT MAKE A SUDDEN CHANGE TO THE OPPOSITE DIRECTION
- TURN LIMITS MUST NOT BE MORE THAN 120 DEGREES RIGHT AND LEFT FROM THE CENTER.

IF YOU DO NOT OBEY THESE PRECAUTIONS, DAMAGE TO THE EQUIPMENT CAN OCCUR

**WARNING:** BEFORE YOU CONNECT THE TOWBAR TO THE AIRCRAFT MAKE SURE YOU SET THE NOSEWHEEL STEERING TO THE OFF POSITION. IT MUST STAY IN THE OFF POSITION WHILE YOU TOW THE AIRCRAFT. THE NOSEWHEEL HAS TO BE IN THE MIDDLE POSITION BEFORE YOU DISCONNECT THE TOWBAR. IF YOU DO NOT DO THIS, YOU CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE EQUIPMENT

(9) Set the nosewheel STEERING switch to the OFF position.

(10) Connect the towbar head (2) to the towing spools (5).

(11) Connect the towbar eye to the tow vehicle.

**NOTE:** Make sure that you engage the lock handle on the towhead.

(12) Remove the chocks from the landing gear wheels.

(13) Set the parking brake to the OFF position.

(14) Slowly tow the aircraft from its parked position.

(15) Do not tow the aircraft at more than a maximum speed of 5 mph (8 kph) around corners or 15 mph (24 kph) in a straight line.

(16) Keep the aircraft turning radii in the limits. The towing turn limits are 120 degrees each side of center. Painted arrows on the two aft NLG doors show these limits.

(17) Engage the parking brake after you tow the aircraft.

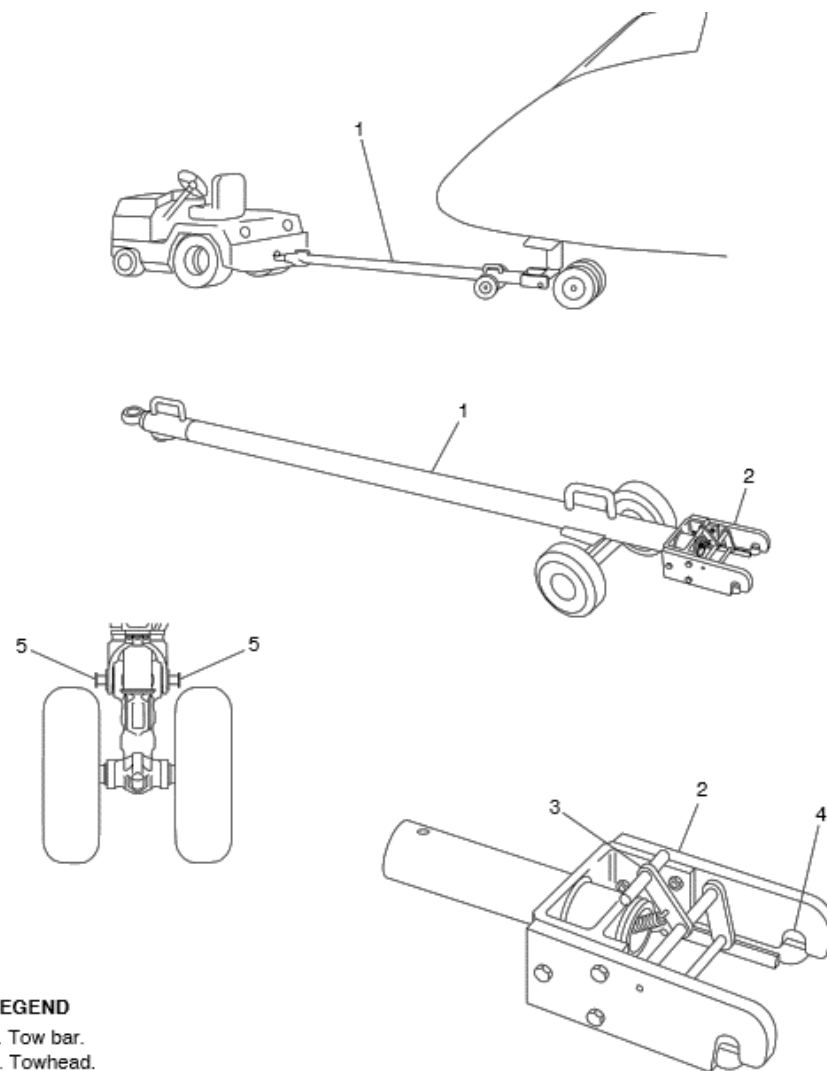
**WARNING:** REMOVE THE TOWBAR FROM THE NOSE LANDING GEAR BEFORE YOU SET THE STEERING SWITCH TO THE ON POSITION. IF YOU SET THE STEERING SWITCH TO THE ON POSITION, THE NOSEWHEELS WILL AUTOMATICALLY GO TO THE NEUTRAL POSITION. IF YOU DO NOT REMOVE THE TOWBAR, YOU CAN CAUSE INJURY TO PERSONS OR DAMAGE TO THE EQUIPMENT

(18) Make sure the nosewheel is straight before you disconnect the towbar.

(19) Disconnect the towbar eye from the tow vehicle.

(20) Disconnect the towbar head (2) from the towing spools (5).

(21) Park the aircraft.



#### LEGEND

1. Tow bar.
2. Towhead.
3. Locking handle.
4. Tow spool attachment point.
5. Towing spool.

Figure 09-01 Tow Bar

## Towing/Moving Aircraft with flat tires

An aircraft with a flat tire or a combination of flat tires on the runway, taxiway or apron is allowed to be moved with the use of tow bar or using suitable rope to clear the runway or taxiway. The maximum combination of flat tires is defined in Figure 09-02.

Make sure that the rims of the wheel with flat tires are not damage and the aircraft is still supported by its own main and nose landing gears.

Obey the precautions that follow when towing an aircraft with two flat tires on one or more axles:

- Use rope on both main landing gears for towing.
- Limit towing speed to 2 mph (3 kph).
- Towing distance should be kept to a minimum to clear the runway
- Avoid sharp turns, abrupt starts and stops.

Refer to the table that follows for maximum allowable towing loads with flat tires:

| Maximum Allowable Towing load for<br>ONE deflated tire (lb) | Maximum Allowable Towing load for<br>TWO deflated tires (lb) |
|-------------------------------------------------------------|--------------------------------------------------------------|
| 7185                                                        | 5987                                                         |

### NOTES (refer figure 09-2):

1. Avoid sharp turns, abrupt starts and stops.
2. Maximum taxiing or towing speed = 5 mph (8 kmh).
3. Maximum taxiing or towing speed = 2 mph (3 kmh).
4. After clearing runway, or if additional tire fails, aircraft should be stopped and serviceable wheel/tire assembly(ies) installed to satisfy conditions in Item No. 2 or 6.

5. Taxiing or towing with two flat tires on same gear can result in wheel damage.

6. After any tire failure or excessive heat condition, the affected wheel assembly must be inspected per applicable Vendor Manual prior to further use.

7. Under multiple failed tire condition, the affected landing gear assemblies and linkages must be inspected for possible structural damage.

8. Towbar required.

9. Use of rope on both main landing gear required.

| ITEM NO. | MAIN GEAR CONDITION                | TIRE FOOTPRINT EXAMPLES | IS TAXIING ALLOWED? | IS TOWING ALLOWED? | DISTANCE ALLOWED (TAXIING & TOWING) | NOSE WHEEL ANGLE OF TURN | REMARKS                                    |
|----------|------------------------------------|-------------------------|---------------------|--------------------|-------------------------------------|--------------------------|--------------------------------------------|
| 1        | ONLY ONE FLAT TIRE (ANY TIRE)      |                         | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 2., 8., 8.                   |
| 2        | TWO FLAT TIRES (ONE ON EACH AXLE)  |                         | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 3., 4., 8., 8.               |
| 3        | TWO FLAT TIRES (ON ONE AXLE)       |                         | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 8., 7., 8.       |
| 4        | THREE FLAT TIRES (ANY COMBINATION) |                         | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 8., 7., 8.       |
| 5        | FOUR FLAT TIRES                    |                         | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 8., 7., 8.       |
|          | NOSE GEAR CONDITION                |                         |                     |                    |                                     |                          |                                            |
| 6        | ONE FLAT TIRE                      |                         | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 2., 4., 8.                   |
| 7        | TWO FLAT TIRES                     |                         | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 6., 7., 8. or 8. |

= FLAT TIRE

Figure 09-02 Towing of the Aircraft (Flat Tire Conditions)

## 09-20-00 Taxiing

The aircraft is moved with engine power, Nosewheel steering, and brakes by a crew member in the pilot seat.

The aircraft is taxied as follows:

- Make sure that the tires are serviceable
- Make sure that the pressure in the parking brake accumulator is a minimum of 1000 psi.
- Set the parking brake lever in the flight compartment to the PARK position.
- Remove the ground wire from the aircraft
- Remove the chokes from the landing gear
- Ensure all external doors of the aircraft are closed
- Make sure that the area around the aircraft is clear of personnel and contamination that may get ingested in the engines
- Start aircraft engines
- Remove all external servicing connections from the aircraft
- Set the Nosewheel STEERING switch to ON
- Set the parking brake lever to the OFF position. Make sure that the caution light for the PARKING BRAKE goes off.
- Get the applicable ground crew and/or ground control clearance to taxi
- With the condition levers set to MAX/1020 rpm, apply sufficient engine power by the POWER levers to start aircraft movement. Decrease the engine power to the DISC position.
- Apply pressure to the brakes to make sure that the brake system is serviceable.
- With the Condition Levers set to MAX/1020 rpm, use the POWER levers to apply sufficient engine power to start aircraft movement. Decrease the engine power to the FLIGHT IDLE position.

**NOTE:** The maximum wind speed for ground maneuvering/static operation and related power setting limitations can be found in the Airplane Flight Manual (PSM1-84-1A), Section 2 – Limitations, Paragraph 2.5.7.

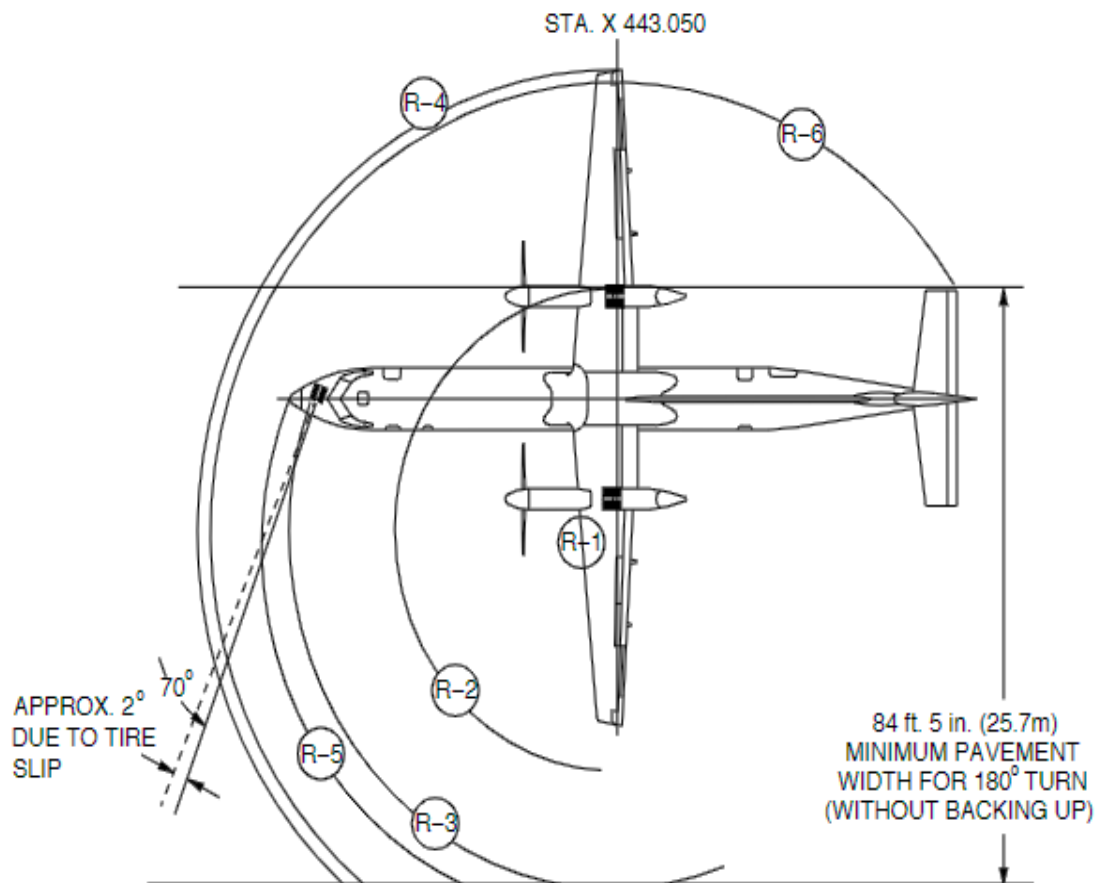
**NOTE:** The nose wheel steering control gives nose wheel steering control of 70 degrees each side of center when you taxi the aircraft in the low speed taxi mode.

**CAUTION:** OBEY THE PRECAUTIONS THAT FOLLOW WHEN YOU TAXI THE AIRCRAFT:

- DO NOT USE THE AIRCRAFT BRAKES UNLESS NECESSARY
- MAKE TURNS LARGE WHEN POSSIBLE.
- MAKE ALL CHANGES TO SPEEDS OR DIRECTIONS SLOWLY WHEN YOU TAXI.
- THE TURN LIMITS MUST NOT BE MORE THAN 120 DEGREES RIGHT AND LEFT FROM THE CENTER WHEN YOU TAXI (FREE CASTERING MODE).

IF YOU DO NOT OBEY THESE PRECAUTIONS, DAMAGE TO THE EQUIPMENT CAN OCCUR

- The aircraft is taxied to the necessary area. There are three taxi modes as follows:
  - (a) The **high speed taxi mode** gives a nose wheel steering angle of 8 degrees each side of center. Control the aircraft with the rudder pedals when you are in the high speed taxi mode.
  - (b) The **low speed taxi mode** gives a nose wheel steering angle of 70 degrees each side of center. Control the aircraft with the nose wheel steering hand–control on the pilot’s side console.
  - (c) The **free castering mode** gives a nose wheel steering angle of 120 degrees each side of center. Control the aircraft with differential thrust and/or brakes.
- When you complete the taxi operation, decrease the engine power to the DISC position and stop the aircraft in a straight line. Make sure that the nose wheel is straight.

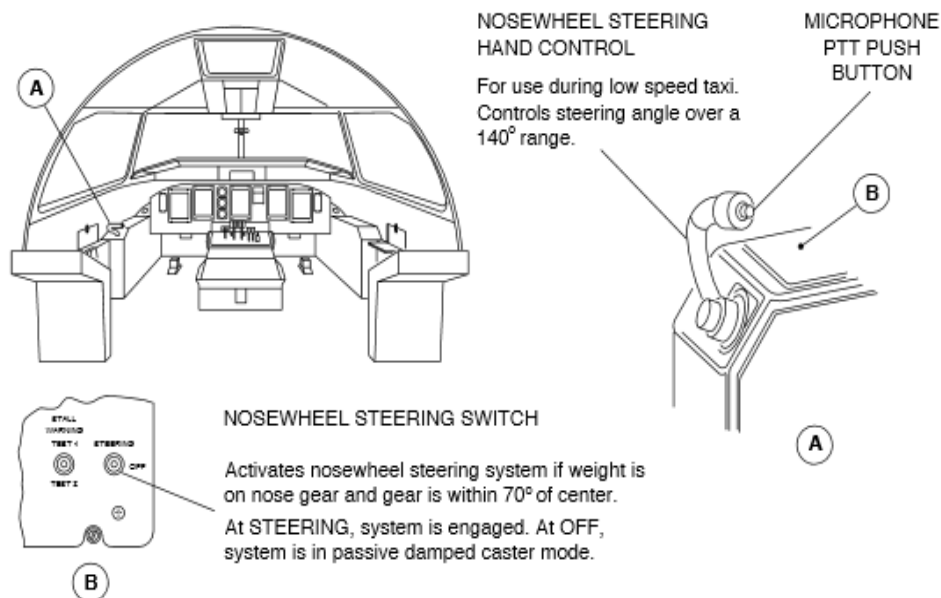


| ITEM             | RADIUS                  |
|------------------|-------------------------|
| R-1 INNER GEAR   | 4 ft. 9 in. (1.52 m)    |
| R-2 OUTER GEAR   | 33 ft. 10 in. (10.32 m) |
| R-3 NOSE GEAR    | 50 ft. 7 in. (15.41 m)  |
| R-4 WING TIP     | 64 ft. 9 in. (19.74 m)  |
| R-5 NOSE         | 54 ft. 10 in. (16.73 m) |
| R-6 ELEVATOR TIP | 62 ft. 9 in. (19.13 m)  |

#### NOTES

1. Dimensions noted are for dry, hard, level surfaces at these tire pressures : 220 psi (loaded), 32 x 8.8-16 standard main-wheel tires, and 87 psi (loaded) for 22 x 6.50-10 standard nose-wheel tires.
2. Nose gear steering limit is approximately 70° left and right.
3. Slip angle of 2° is approximate only and may vary with aircraft configuration, loading and tire wear.
4. Dimensions given for maneuvering clearance and turning radii are minimum recommended limits.
5. Tire pressures shown are for calculation purposes only. Refer to AMM Ch. 12 for service pressures.

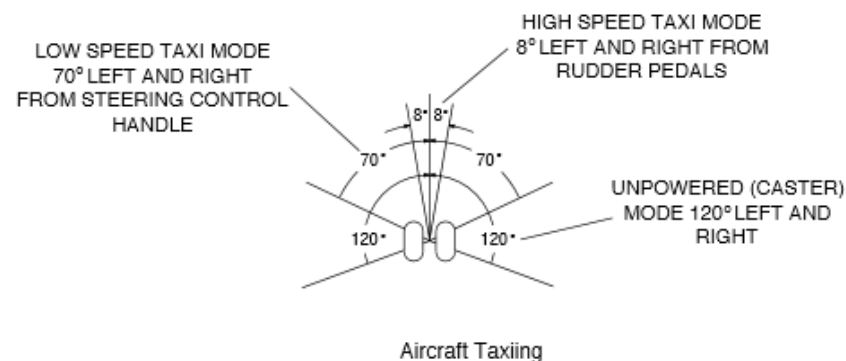
**Figure 09-03 Aircraft Taxiing Turning Radii with Minimum Power**



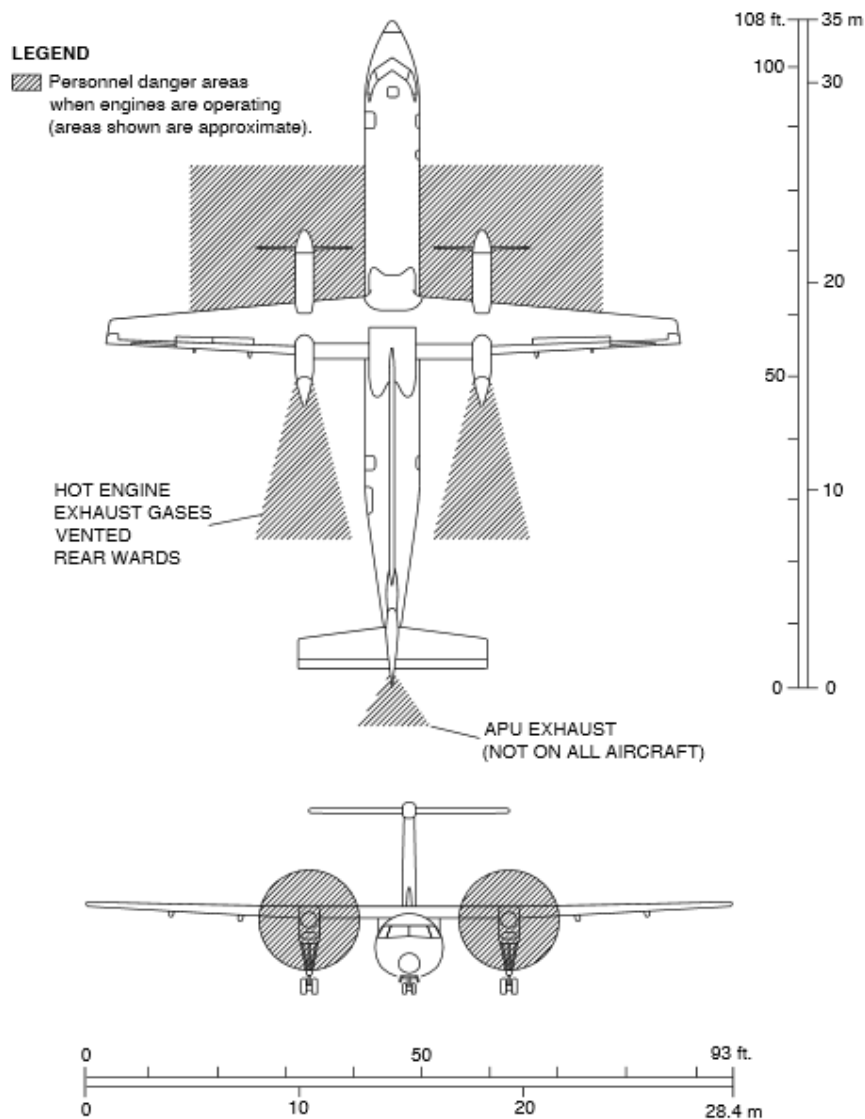
THE NOSEWHEEL STEERING SYSTEM INCORPORATES THREE MODES OF OPERATION FOR TAXIING:

1. A HIGH SPEED MODE, CONTROLLED BY THE RUDDER PEDALS, GIVING A STEERING ANGLE OF 8 DEGREES EACH SIDE OF CENTER.
2. A LOW SPEED MODE, CONTROLLED BY A STEERING HAND CONTROL FOUND ON THE PILOT'S SIDE CONSOLE, GIVING A STEERING ANGLE OF 70 DEGREES EACH SIDE OF CENTER.
3. A FREE CASTERING (UNPOWERED) MODE, CONTROLLED BY DIFFERENTIAL ENGINE THRUST AND/OR BRAKING GIVING A STEERING ANGLE OF 120 DEGREES EACH SIDE OF CENTER.

**Figure 09-04 Steering Instructions**

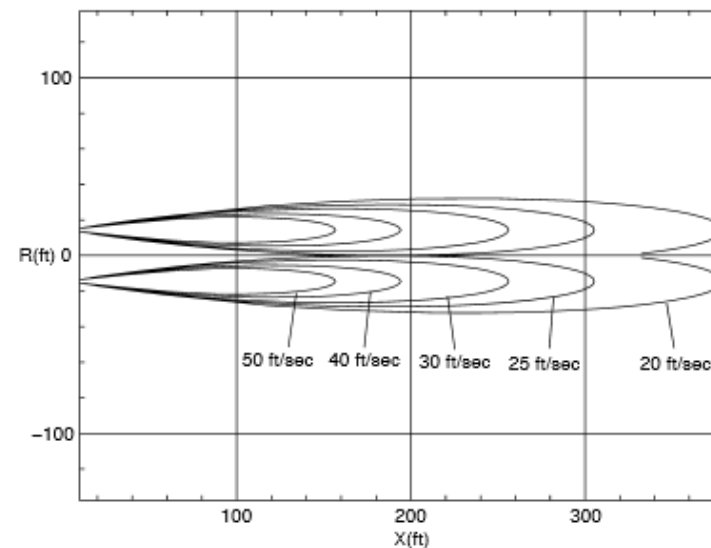


**Figure 09-05 Steering Limits & Turning Radii**



Propeller / Engine Exhaust Danger Areas  
Figure 09-06 Danger Areas

SLIPSTREAM VELOCITY CONTOURS (~18% TORQUE) (STATIONARY ON GROUND)



SLIPSTREAM VELOCITY CONTOURS (~43% TORQUE) (STATIONARY ON GROUND)

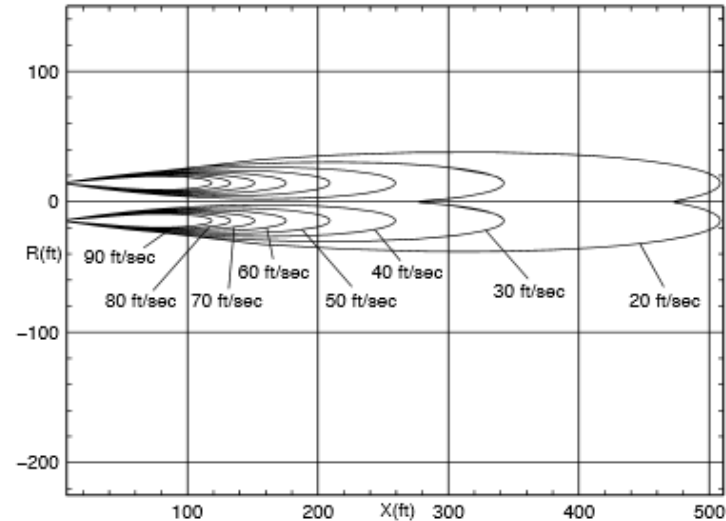


Figure 09-07 Exhaust Danger Trail



| ITEM NO. | MAIN GEAR CONDITION                | TIRE FOOTPRINT EXAMPLES                                                           | IS TAXIING ALLOWED? | IS TOWING ALLOWED? | DISTANCE ALLOWED (TAXIING & TOWING) | NOSE WHEEL ANGLE OF TURN | REMARKS                                    |
|----------|------------------------------------|-----------------------------------------------------------------------------------|---------------------|--------------------|-------------------------------------|--------------------------|--------------------------------------------|
| 1        | ONLY ONE FLAT TIRE (ANY TIRE)      |  | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 2., 6., 8.                   |
| 2        | TWO FLAT TIRES (ONE ON EACH AXLE)  |  | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 3., 4., 6., 8.               |
| 3        | TWO FLAT TIRES (ON ONE AXLE)       |  | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 6., 7., 9.       |
| 4        | THREE FLAT TIRES (ANY COMBINATION) |  | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 6., 7., 9.       |
| 5        | FOUR FLAT TIRES                    |  | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 6., 7., 9.       |
|          | NOSE GEAR CONDITION                |                                                                                   |                     |                    |                                     |                          |                                            |
| 6        | ONE FLAT TIRE                      |  | YES                 | YES                | UNLIMITED                           | NORMAL                   | SEE NOTES 1., 2., 4., 8.                   |
| 7        | TWO FLAT TIRES                     |  | YES                 | YES                | MINIMUM TO CLEAR RUNWAY             | 10° MAXIMUM              | SEE NOTES 1., 3., 4., 5., 6., 7., 8. or 9. |

 = FLAT TIRE

## NOTES

1. Avoid sharp turns, abrupt starts and stops.
2. Maximum taxiing or towing speed = 5 mph (8 kmh).
3. Maximum taxiing or towing speed = 2 mph (3 kmh).
4. After clearing runway, or if additional tire fails, aircraft should be stopped and serviceable wheel/tire assembly(ies) installed to satisfy conditions in Item No. 2 or 6.
5. Taxiing or towing with two flat tires on same gear can result in wheel damage.
6. After any tire failure or excessive heat condition, the affected wheel assembly must be inspected per applicable Vendor Manual prior to further use.
7. Under multiple failed tire condition, the affected landing gear assemblies and linkages must be inspected for possible structural damage.
8. Towbar required.
9. Use of rope on both main landing gear required.

**Figure 09-08 Taxiing of the Aircraft (Flat Tire Conditions)**



# ATA-10

## PARKING AND MOORING

## 10-00-00 Parking and Mooring

De Havilland Aircraft of Canada Limited (De Havilland Canada) holds the position that while the aircraft is in a parked condition, the aircraft must be parked so that safety is the prime concern. De Havilland Canada realizes that conditions for parking the aircraft may vary, and recommends that at the operator's discretion the operator may choose to do all of the procedures listed in the TASK that follows, or the parts thereof that the operator deems sufficient to provide complete safety in the event that a fast turn-around is necessary.

The maintenance procedure that follows gives instructions to park the aircraft. It also gives instructions for the protection of the aircraft from environmental damage when the aircraft is parked for a short time (seven days or less).

If the aircraft will be parked for a period of eight days or more and you do not wish to extend the parking period, do the applicable storage procedures.

The steps that follow are done at all times when the aircraft is parked in or out of the hangar:

- Set the parking brake lever to the PARK position.
- Install the cover on the center console.
- Engage the ground lock for the nose landing gear
- Install the lockpins in the main landing gear
- Set the CONTROL LOCK lever on the flight control pedestal to the ON position.
- Ground the aircraft.
- Install the wheel chocks at the nose landing gear and at the main landing gear.

When the aircraft is parked outdoors, do the steps that follow:

- Point the aircraft nose into the wind.
- Install the engine intake and exhaust covers.

- Install the wheel chocks at the nose landing gear and at the main landing gear.

**WARNING:** DO NOT INSTALL THE COVERS WHEN THE PITOT STATIC PROBES ARE HOT. THE COVER WILL MELT AND CAUSE A BLOCKAGE OF THE HOLES IN THE PITOT STATIC PROBE. THIS CAN CAUSE INJURY TO PERSONNEL AND/OR DAMAGE TO EQUIPMENT.

- Install the covers for the Pitot-static probe, as necessary.
- Install the cover for the angle-of-attack vane.
- Install the propeller restraints.
- Install the wheel covers on the nose landing gear and the main landing gear as necessary.
- Install all other supplied covers and plugs as necessary.
- Seal the center over wing fuselage to wing fairing access panels with aluminum foil tape over the quick release fasteners (panels 261LT, 261HT, 261FT, 261CT, 262HT, 262FT, 262AT, 262JT, 262GT, and 262BT)

If the aircraft was parked outside with snow accumulation and/or cold temperatures refer to Ramp Servicing Manual, PSM 1-84-2S, Chapter 6, Cold Weather Servicing.

Refer to Figure 10-01: **Engaging the Nose Gear Ground Lock** The nose gear ground lock is engaged as follows:

**WARNING:** DO NOT USE FORCE WHEN YOU ENGAGE THE GROUND LOCK OF THE NOSE LANDING GEAR. IF FORCE IS NECESSARY, THE LANDING GEAR MAY NOT BE SAFE AND CAN RETRACT ACCIDENTALLY. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE AIRCRAFT.

1. The pushbutton is pushed in to release the door of the NLG ground lock
2. The door of the nose gear ground lock is pulled away from the fuselage
3. The door of the nose gear ground lock is turned 90 degrees clockwise to engage the ground lock.

If you cannot engage the NLG ground lock correctly, do the steps that follow before you examine the nose landing gear:

NOTE: If you cannot engage the NLG ground lock correctly, the nose landing gear is not safe.

(a) Install a jack adapter at the forward fuselage jacking point.

(b) Put a jack below the forward fuselage jacking point.

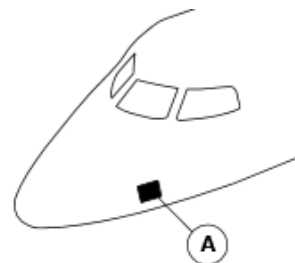
**WARNING:** DO NOT LIFT THE AIRCRAFT ON JACKS IF YOU THINK THE LOAD AT THE NOSE-FUSELAGE JACKING POINT WILL BE 4700 LB (2132 KG) OR MORE. IF YOU DO, YOU CAN CAUSE DAMAGE TO THE AIRCRAFT STRUCTURE. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO THE AIRCRAFT.

c) Extend the jack sufficiently to remove the weight of the aircraft from the nose landing gear.

NOTE: Two crew members are necessary to set up the jack. One member operates the jack and the other holds the jack adapter in the receptacle until the jack correctly engages the adapter.

(d) Tighten the lock collar on the jack to lock the jack extension.

NOTE: You can remove the nose fuselage jack and adapter when the aircraft is serviceable



#### LEGEND

1. Release button.
2. Nose gear ground lock door.



(A) **DISENGAGED**  
(FULLY RETRACTED)



(A) **ENGAGED**  
(FULLY EXTENDED)

**Figure 10-01 Nose Gear Ground Lock**

## Refer to Figure 10-02: Installation of the Main Landing Gear (MLG) Lock Pins

The MLG lockpins are installed in the lockpin holes of the stabilizer down-lock mechanism.

If the MLG lockpin cannot be put in its hole easily, install a jack adapter and put a jack below the applicable wing jacking point. The jack can be extended sufficiently to remove the weight of the aircraft from the main landing gear.

(1) Install the MLG lockpin (2) in the lockpin hole (1) of the stabilizer down-lock mechanism.

(2) If you cannot put the MLG lockpin in its hole easily, the main landing gear is not safe. Do the steps that follow before you examine the main landing gear:

(a) Install a jack adapter.

(b) Put a jack below the applicable wing jacking point.

(c) Extend the jack sufficiently to remove the weight of the aircraft from the main landing gear.

**NOTE:** Two crew members are necessary to set up the jack. One member operates the jack and the other holds the jack adapter in the receptacle until the jack touches the adapter.

(d) Tighten the lock collar on the jack to lock the jack extension.

**NOTE:** You can remove the wing jack and the adapter when the aircraft is serviceable

**WARNING:** DO NOT USE FORCE WHEN YOU INSTALL OR REMOVE THE LOCKPIN. IF FORCE IS NECESSARY, THE LANDING GEAR MAY NOT BE SAFE AND CAN RETRACT ACCIDENTALLY. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO THE AIRCRAFT

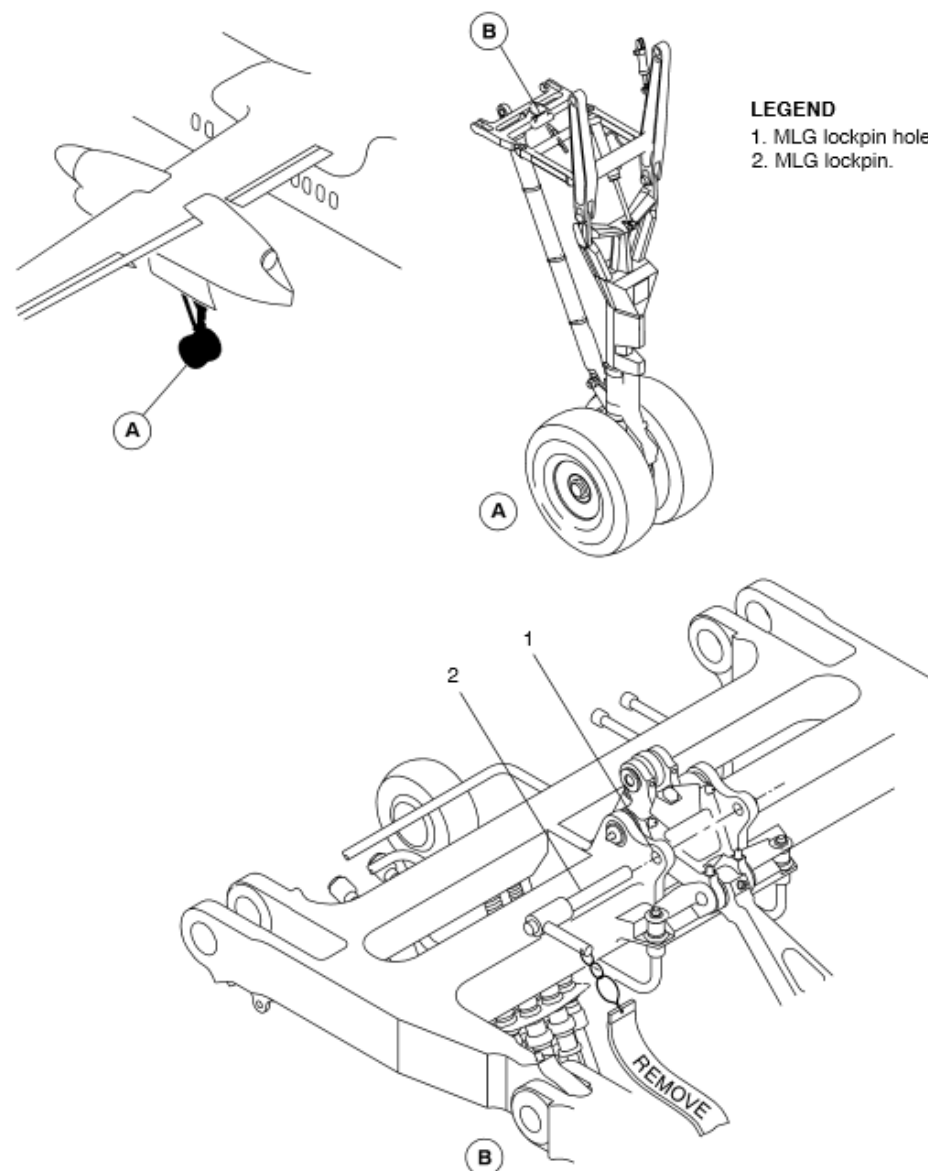


Figure 10-02 MLG Lock Pin Installation

### Refer to Figure 10-03: Static Grounding of the Aircraft

There is a ground point on the drag brace of each main landing gear assembly. There is no ground point on the nose landing gear.

Electrical bonding between the aircraft and the refueling vehicle is mandatory during the refuel/defuel operations.

The static grounding of the aircraft is mandatory when you do work on or near the oxygen system and its components.

For aircraft in the hangar, the static grounding of the aircraft is mandatory when you do maintenance work with equipment powered from the main electrical grid.

This includes:

- Electrically powered tools,
- Electrically powered work lights,
- Electrically powered instruments,
- Other electrically powered equipment (which includes aircraft power derived from the power grid)

For aircraft in the hangar, the static grounding of the aircraft is mandatory if fuel is or was in the fuel tanks.

For aircraft in the hangar, the static grounding of the aircraft is mandatory in flammable conditions. This includes painting and solvent cleaning.

Since several activities may be occurring on the aircraft at once, De Havilland recommends the static grounding of the aircraft at all times when in the hangar.

For aircraft outside of the hangar, the static grounding of the aircraft is mandatory when you do maintenance operations with thunderstorms in the area.

For aircraft outside of the hangar, De Havilland recommends the static grounding of the aircraft when you do maintenance work with the aircraft and equipment powered as follows:

- Aircraft electrical system: APU, engine generators or battery; or
- An external power source such as a mobile self-contained power source (ground power cart) not connected to the main electrical grid; or
- Battery.

For aircraft outside the hangar, De Havilland recommends the static grounding of the aircraft when you do checks with the external fuel drain valve and the magnetic dipstick.

The static grounding of the aircraft is not necessary for a turnaround flight and no maintenance being done.

De Havilland recommends the static grounding of the aircraft for short-term parking and storage and when no maintenance being done.

**WARNING:** GROUND THE AIRCRAFT AND THE WORKSTAND BEFORE YOU DO FUEL SYSTEM MAINTENANCE. AN ELECTROSTATIC SPARK CAN CAUSE AN EXPLOSION OR A FIRE.

**WARNING:** GROUND THE AIRCRAFT WHEN YOU DO WORK ON OR NEAR THE OXYGEN SYSTEM AND ITS COMPONENTS. IF YOU DO NOT DO THIS, YOU CAN CAUSE A FIRE OR AN EXPLOSION.

Do the steps that follow to ground the aircraft:

- Attach one end of the grounding cable (2) to a correct ground point in a hangar or the ramp area,
- Attach the alligator clamp of the grounding cable (2) to the ground stud (1) on the applicable main landing gear drag brace (3).

NOTE: If for any reason the MLGs are retracted the aircraft may be grounded through the alternate grounding points (refer to Airport Planning Manual, chapter 5) that follow:

- If the MLG are retracted but the MLG door is open, the aircraft should be



grounded through the ground stud (1),

- Alternately ground the aircraft near the grounding point at the pressure refuelling panel, or
- Ground the aircraft at ground points over the left or right wing

**CAUTION:** WHEN YOU STATIC GROUND AN AIRCRAFT, FIRST ATTACH ONE END OF THE GROUND CABLE TO A CORRECT GROUND POINT IN THE HANGAR OR THE RAMP AREA. THEN ATTACH THE OTHER END OF THE GROUND CABLE TO THE AIRCRAFT GROUND POINT. IF YOU DO NOT DO THIS, YOU COULD RECEIVE AN ELECTROSTATIC SHOCK

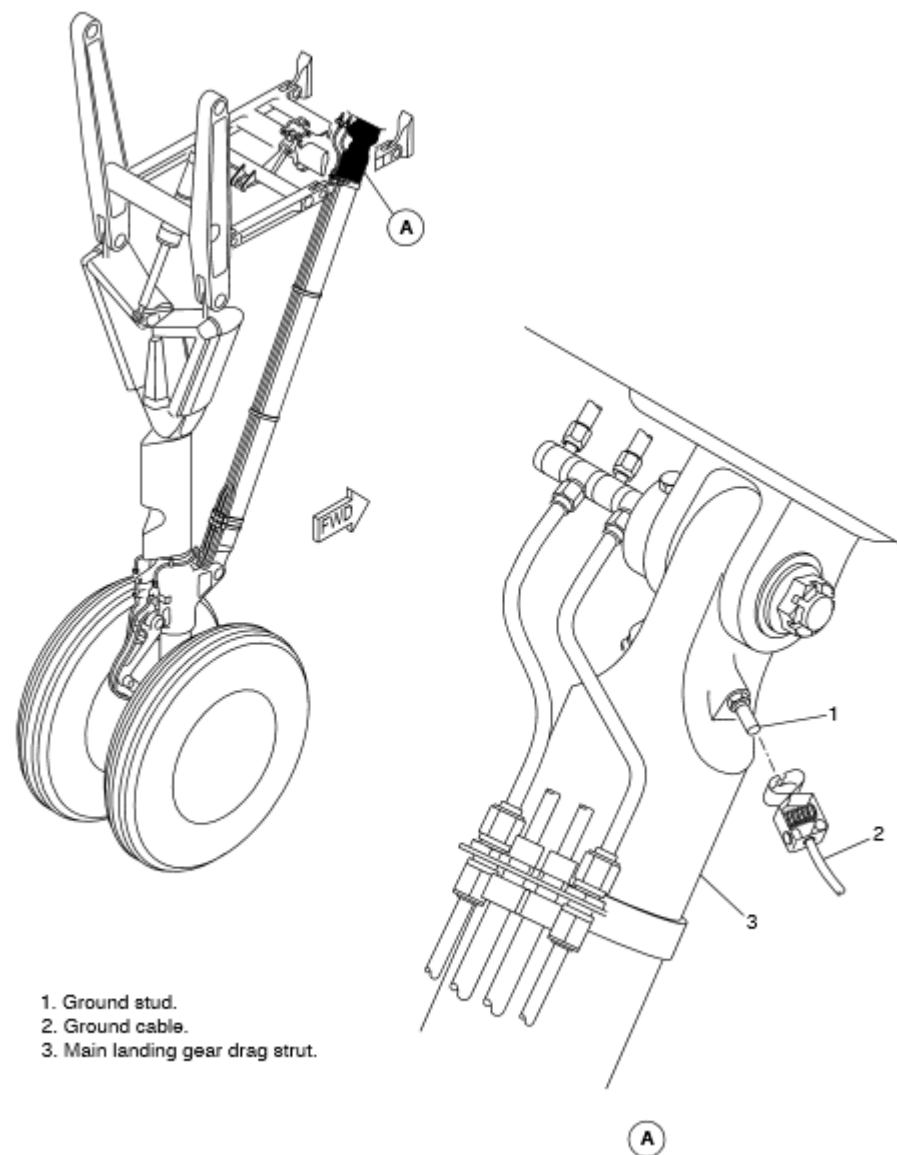


Figure 10-03 Aircraft Grounding Point



## Mooring

Mooring of the aircraft is done to secure the aircraft to the ground while the aircraft is in a parked position. Mooring is done when the aircraft is parked outside during high wind conditions or when the aircraft is parked on snow or ice covered surfaces. Rope and ground attachment rings are used to moor the aircraft at the nose landing gear and at both the main landing gears.

De Havilland Aircraft of Canada Limited (De Havilland Canada) recommends that the aircraft must be moored after being placed in the parked condition outside or relocated to a protected location for the following weather conditions :

- When it is parked on dry concrete or asphalt and the winds are, or are forecast to be, 63 mph (102 km/h = 54.75 Knots) or more.
- When it is parked on wet concrete or asphalt and the winds are, or are forecast to be, 46 mph (74 km/h = 40 knots) or more.
- When it is parked on snow or ice covered surfaces.

**NOTE:** To reduce any pitching tendencies during high winds, it is recommended that the aircraft flaps are set to 0 degrees and the aircraft be fully fueled and the center of gravity be moved as far forward as possible.

### Mooring at the Nose Landing Gear

Refer to Figure 10-04.

Moor the aircraft at the nose landing gear as follows:

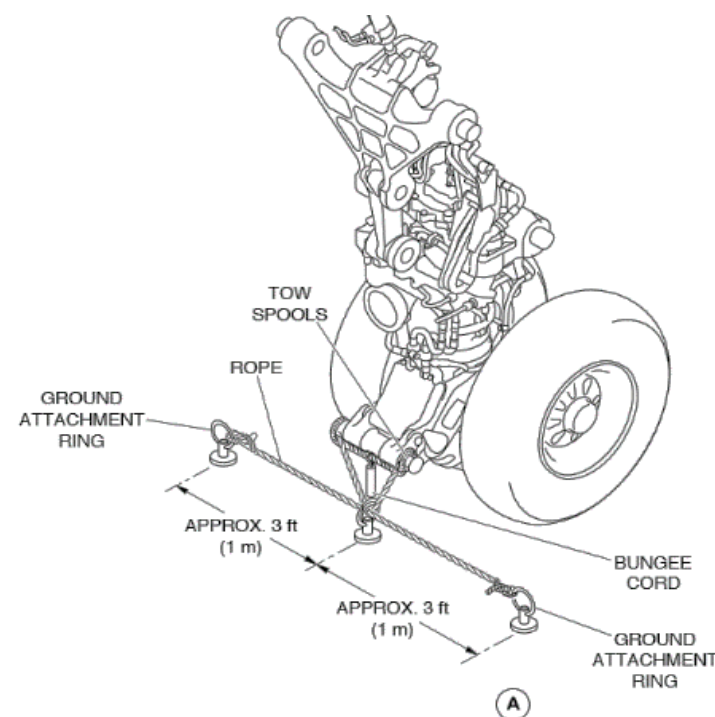
(1) Make a loop with the rope. Put the rope around the tow spools of the nosewheel and through the ground attachment rings.

**NOTE:** With the main landing gear chocks put in their correct positions, you

can move the chock away from the front of the tire at the nose landing gear. This will help you to get access to moor the nose landing gear to its ground attachment rings.

(2) While you keep tension in the ropes, attach them to the ground attachment rings. Make sure that the ropes stay tight.

(3) Install a bungee cord between the rope and the middle ground attachment ring. Make sure that the bungee cord has tension.



**Figure 10-04 Mooring at NLG**

## Mooring at the Main Landing Gear

Refer to Figure 10-05.

Moor the aircraft at the Main landing gear as follows:

- (1) Install a protective spool (1) at each main landing gear shock strut outer web.
- (2) Make a double loop with the rope (2) around the protective spool (1) on the outer web of the main landing gear and through the ground attachment rings (3).
- (3) While you keep tension on the ropes (2), attach them to the ground attachment rings (3). Make sure that the ropes (2) stay tight.

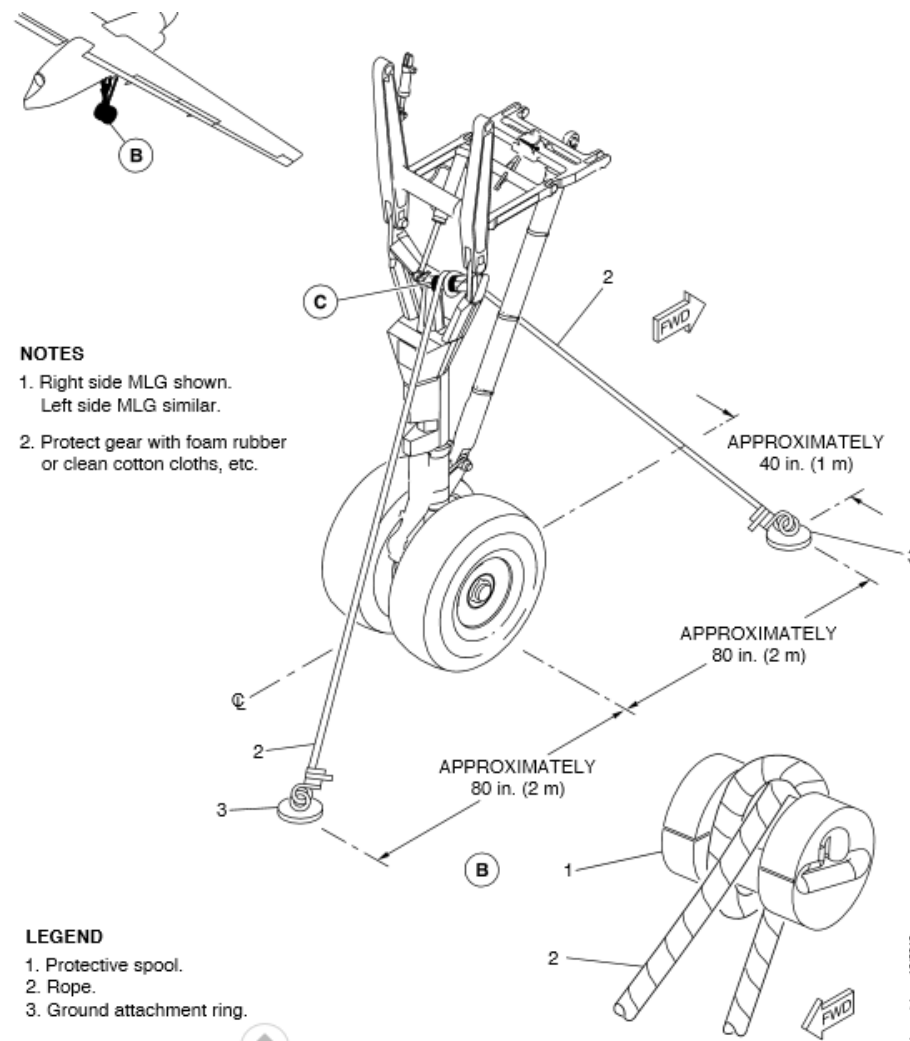


Figure 10-05 Mooring at MLG



# ATA-11

## PLACARD AND MARKING

## 11-00-00 Placards and Markings

De Havilland Aircraft of Canada Limited (De Havilland Canada) considers every placard, label and marking to be important. Some placards, labels and markings are mandatory installations as determined by local government regulations and local airworthiness regulations.

### A. Exterior Placards and Markings

(1) For all the illustrations and locations of each of the placards, labels and markings that are installed on the exterior of the aircraft, Refer to PSM 1-84-4, the Aircraft Illustrated Parts Catalog (AIPC) Chapter 11 for details.

### B. Interior Placards and Markings

(1) For all the illustrations and locations of each of the placards, labels and markings that are installed on the interior of the aircraft, Refer to PSM 1-84-4, the Aircraft Illustrated Parts Catalog (AIPC) Chapter 11 for details.

The aircraft has placards, labels, and markings.

The aircraft has exterior placards and markings on the following:

- Fuselage
- Landing Gear
- Wing
- Nacelles and Propellers

The aircraft also has interior placards and markings on the following:

- Flight Compartment
- Lavatory
- Cabin
- Galley
- Baggage Compartments



# ATA-12

## SERVICING

## 12-00-00 Servicing

The servicing of the systems is covered under respective ATA Chapters.